



Second International Symposium on Marine Propulsors 2011

Workshop: Propeller Performance

Potsdam Propeller Test Case (PPTC)

Open Water Test with the Model Propeller VP1304

Case 2.1

Potsdam, May 2011

Case 2.1

Open Water Tests

Test Description

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1 Introduction

In the intention to offer research groups the possibility to test and validate their numerical tools the SVA Potsdam GmbH (SVA) investigated the controllable pitch propeller VP1304. The test results are published in course of the propeller workshop of the Second International Symposium on Marine Propulsors 2011 (smp'11). The geometry and results of the propeller are presented by the name PPTC, an acronym for "Potsdam Propeller Test Case".

In the following the measurements of the open water tests and the comparison between measurements and calculations are presented.

A more detailed description of the open water tests conducted at the towing tank of the SVA is presented in the SVA report 3752 [1], which can be found on the website sva-potsdam.de.

2 Model propeller VP1304

The propeller is a controllable pitch propeller. This affects the propeller blade design near the hub and results in a 0.3 mm gap between hub and propeller blade near the leading and trailing edge of the propeller.

The hub cap and aft fairing were designed according to the 22nd ITTC (1999) proceedings. The gap between aft fairing and hub has a width of 3 mm. Also a dummy hub, having the same dimensions and mass as the propeller hub, was manufactured.

In Table 1 the main data of the propeller is given.

Table 1: Main data of model propeller VP1304

VP1304			
Diameter	D	[m]	0.250
Pitch ratio $r/R = 0.7$	$P_{0.7}/D$	[-]	1.635
Area ratio	A_E/A_0	[-]	0.77896
Chord length $r/R = 0.7$	$c_{0.7}$	[m]	0.10417
Skew	θ_{EXT}	[°]	18.837
Hub ratio	d_h/D	[-]	0.300
Number of blades	Z	[-]	5
Sense of rotation		[-]	right
Type		[-]	controllable pitch propeller

3 Open water tests

The propeller is investigated in a pull configuration in the towing tank of the SVA. For the open water tests the dynamometer H39 from Kempf & Remmers has been used. During the tests the propeller shaft is submerged by $1.5 D$.

Measurements with a dummy hub were conducted, in order to get data of the idle torque, the force in the gap between the hub and the aft fairing and the resistance of the hub cap.

The open water tests were carried out at two different numbers of revolutions to evaluate the dependency on the Reynolds number. However for the evaluation of the computational results solely the tests with $n = 15 \text{ s}^{-1}$ are taken into account.

Table 2: Details of model tests

PPTC	
Towing tank	
Dimensions of towing tank	280 x 9 x 4.5 m
Propeller:	VP1304
Propeller type	controllable pitch propeller
Material of model propeller	brass
Diameter of the propeller	0.25 m
Sense of rotation	right (looking on pressure side)
Arrangement:	
Shaft inclination	0°
Dynamometer shaft	behind the propeller
Measuring equipment:	dynamometer H39
	$n_{\max} = 60 \text{ s}^{-1}$
	$T_{\max} = 1 \text{ kN}$
	$Q_{\max} = 50 \text{ Nm}$

4 Requested computations

It was requested to calculate the open water characteristics of the propeller for the advanced coefficients $J = 0.6, 0.8, 1.0, 1.2$ and 1.4 . The computations should be conducted according to the operation conditions given in Table 4.

Table 4: Operation conditions

Water density	$(t_W = 17.5 \text{ }^\circ\text{C})$	ρ	[kg/m ³]	998.67
Kinematic viscosity	$(t_W = 17.5 \text{ }^\circ\text{C})$	ν	[m ² /s]	1.070E-06
Number of revolutions		n	[1/s]	15.00

The thrust and torque of the propeller should be made dimensionless employing the rate of revolution n , the propeller diameter D and the density of water ρ as follows:

Advance coefficient $J = \frac{V_A}{n \cdot D}$

Thrust coefficient $K_T = \frac{T}{\rho \cdot n^2 \cdot D^4}$

Torque coefficient $K_Q = \frac{Q}{\rho \cdot n^2 \cdot D^5}$

Open water efficiency $\eta_0 = \frac{J}{2\pi} \cdot \frac{K_T}{K_Q}$

with V_A being the advance speed, T the thrust and Q the propeller torque

5 Open water test results

The measured values in the open water tests were corrected with the values of the idle torque and the gap force.

The measured and corrected open water characteristics for the five requested advance coefficients are given in Table 4 for the number of revolutions $n = 15 \text{ s}^{-1}$.

Table 4: Measured open water characteristics (by polynomials)

J	K_T	$10K_Q$	η_0
0.60	0.6288	1.3964	0.4300
0.80	0.5100	1.1780	0.5512
1.00	0.3994	0.9749	0.6520
1.20	0.2949	0.7760	0.7258
1.40	0.1878	0.5588	0.7487

6 Reference

- [1] Barkmann, U.
Potsdam Propeller Test Case (PPTC) - Open Water Tests with the Model Propeller VP1304,
Report 3752, Schiffbau-Versuchsanstalt Potsdam, April 2011

Case 2.1

Open Water Tests

Evaluation

7 Evaluation

Calculations from 14 different groups were obtained for the evaluation of the open water characteristics, employing 13 different solvers and submitting 19 calculations, of which are two for the comparison of different turbulence models. In the following all participants and the used solvers are listed:

Group	Solver	Acronym
Berg-Propulsion	OpenFOAM	Berg-OpenFOAM
Cradle	SC/Tetra	Cradle-SC/Tetra
CSSRC	ANSYS Fluent	CSSRC-Fluent
HSVA	PPB	HSVA-PPB
	QCM	HSVA-QCM
INSEAN	PFC	INSEAN-PFC
MARIC	ANSYS Fluent	MARIC-Fluent
SSPA	ANSYS Fluent	SSPA-Fluent
SVA	Vortex	SVA-Vortex
TUHH	FreSCO+	TUHH-FreSCO
University of Genua	Panel	UniGenua-Panel
	OpenFOAM	UniGenua-OpenFOAM
	StarCCM+	UniGenua-StarCCM(kw)
	StarCCM+	UniGenua-StarCCM(ke)
University of Trieste	ANSYS CFX	UniTriest-CFX
VicusDT	StarCCM+	VicusDT-StarCCM
VOITH	Comet	VOITH-Comet
	OpenFOAM	VOITH-OpenFOAM
VTT	FinFlo	VTT-FinFlo

In the following the data is evaluated with respect to the thrust coefficient K_T and torque coefficient $10K_Q$ and the open water efficiency η_O . For each value an absolute (e.g. ΔK_T) and a relative comparison (e.g. $\Delta K_T [\%]$) to the measured values are given. The absolute difference is defined as the difference between the calculated and the measured value, while the relative difference is the same value expressed as percentage with respect to the measured value.

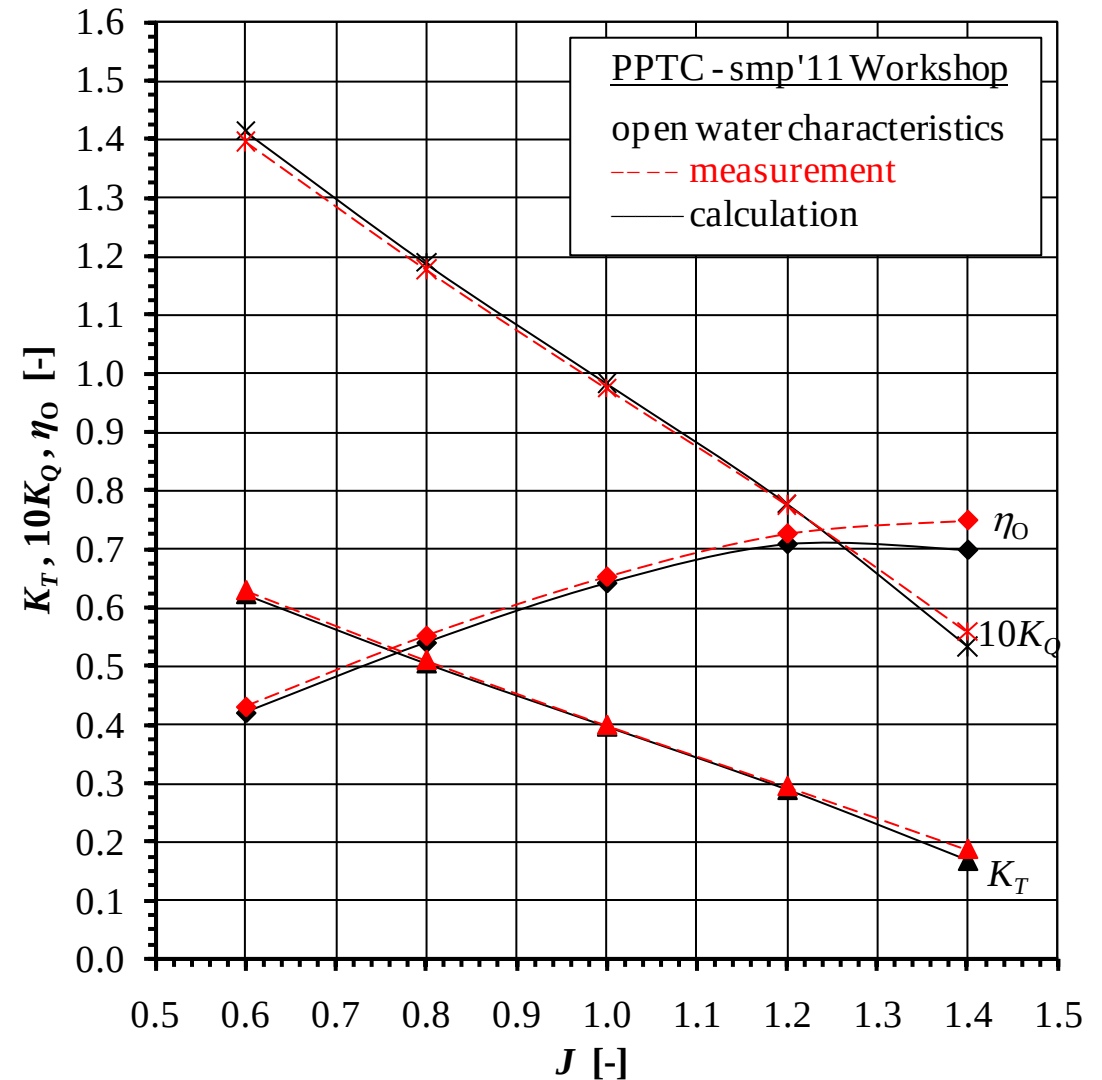
7.1 Berg-OpenFOAM

Characteristic of propeller (model scale)

J	K_T EFD	K_T CFD	ΔK_T	ΔK_T [%]
0.60	0.6288	0.6230	-0.0058	-0.93
0.80	0.5100	0.5050	-0.0050	-0.97
1.00	0.3994	0.3970	-0.0024	-0.59
1.20	0.2949	0.2890	-0.0059	-2.01
1.40	0.1878	0.1680	-0.0198	-10.53

J	$10K_Q$ EFD	$10K_Q$ CFD	$\Delta 10K_Q$	$\Delta 10K_Q$ [%]
0.60	1.3964	1.4150	0.0186	1.33
0.80	1.1780	1.1890	0.0110	0.93
1.00	0.9749	0.9840	0.0091	0.94
1.20	0.7760	0.7780	0.0020	0.25
1.40	0.5588	0.5340	-0.0248	-4.44

J	η_O EFD	η_O CFD	$\Delta \eta_O$	$\Delta \eta_O$ [%]
0.60	0.4300	0.4210	-0.0090	-2.10
0.80	0.5512	0.5410	-0.0102	-1.85
1.00	0.6520	0.6430	-0.0090	-1.38
1.20	0.7258	0.7100	-0.0150	-2.18
1.40	0.7487	0.7000	-0.0487	-6.50



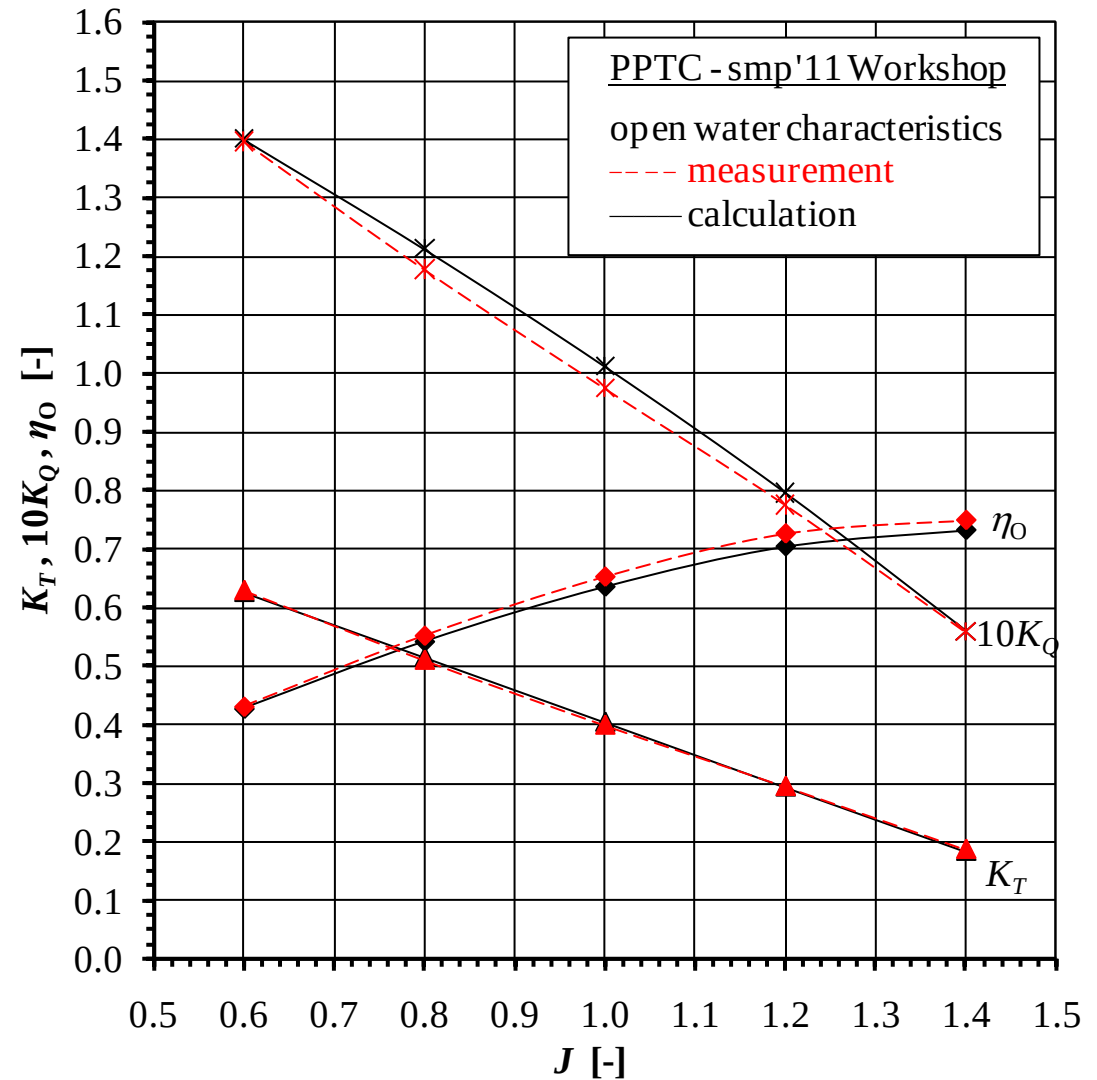
7.2 Cradle-SC/Tetra

Characteristic of propeller (model scale)

J	K_T EFD	K_T CFD	ΔK_T	ΔK_T [%]
0.60	0.6288	0.6270	-0.0018	-0.29
0.80	0.5100	0.5160	0.0060	1.18
1.00	0.3994	0.4050	0.0056	1.41
1.20	0.2949	0.2940	-0.0009	-0.31
1.40	0.1878	0.1840	-0.0038	-2.00

J	$10K_Q$ EFD	$10K_Q$ CFD	$\Delta 10K_Q$	$\Delta 10K_Q$ [%]
0.60	1.3964	1.4010	0.0046	0.33
0.80	1.1780	1.2130	0.0350	2.97
1.00	0.9749	1.0120	0.0371	3.81
1.20	0.7760	0.7960	0.0200	2.57
1.40	0.5588	0.5600	0.0012	0.21

J	η_O EFD	η_O CFD	$\Delta \eta_O$	$\Delta \eta_O$ [%]
0.60	0.4300	0.4270	-0.0030	-0.71
0.80	0.5512	0.5420	-0.0092	-1.67
1.00	0.6520	0.6360	-0.0160	-2.45
1.20	0.7258	0.7050	-0.0208	-2.87
1.40	0.7487	0.7330	-0.0157	-2.09



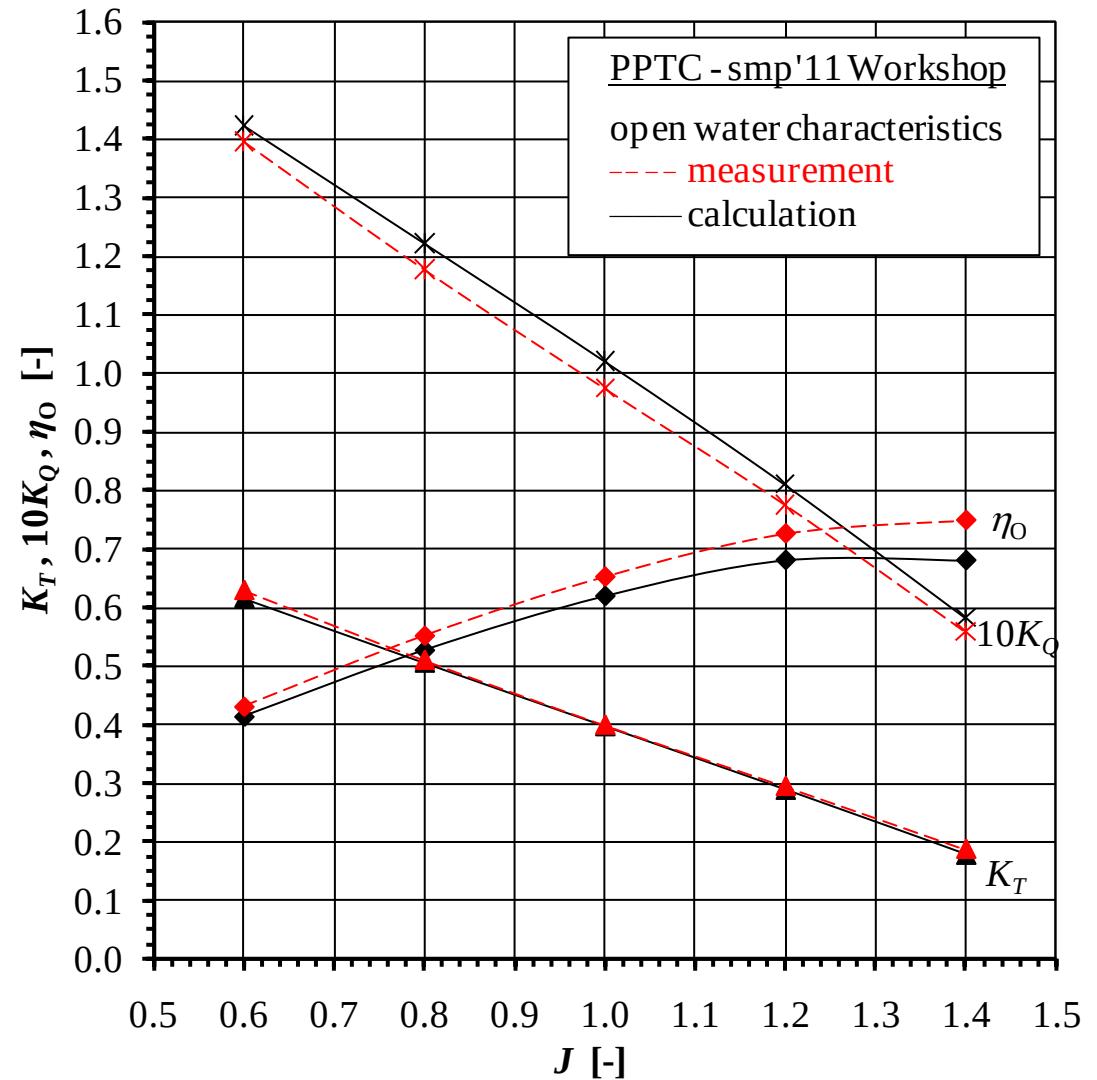
7.3 CSSRC-Fluent

Characteristic of propeller (model scale)

J	K_T EFD	K_T CFD	ΔK_T	ΔK_T [%]
0.60	0.6288	0.6158	-0.0130	-2.07
0.80	0.5100	0.5055	-0.0045	-0.88
1.00	0.3994	0.3971	-0.0023	-0.57
1.20	0.2949	0.2887	-0.0062	-2.11
1.40	0.1878	0.1779	-0.0099	-5.25

J	$10K_Q$ EFD	$10K_Q$ CFD	$\Delta 10K_Q$	$\Delta 10K_Q$ [%]
0.60	1.3964	1.4239	0.0275	1.97
0.80	1.1780	1.2230	0.0450	3.82
1.00	0.9749	1.0215	0.0466	4.78
1.20	0.7760	0.8107	0.0347	4.47
1.40	0.5588	0.5832	0.0244	4.36

J	η_O EFD	η_O CFD	$\Delta \eta_O$	$\Delta \eta_O$ [%]
0.60	0.4300	0.4130	-0.0170	-3.96
0.80	0.5512	0.5263	-0.0249	-4.52
1.00	0.6520	0.6187	-0.0333	-5.11
1.20	0.7258	0.6801	-0.0457	-6.30
1.40	0.7487	0.6797	-0.0690	-9.21



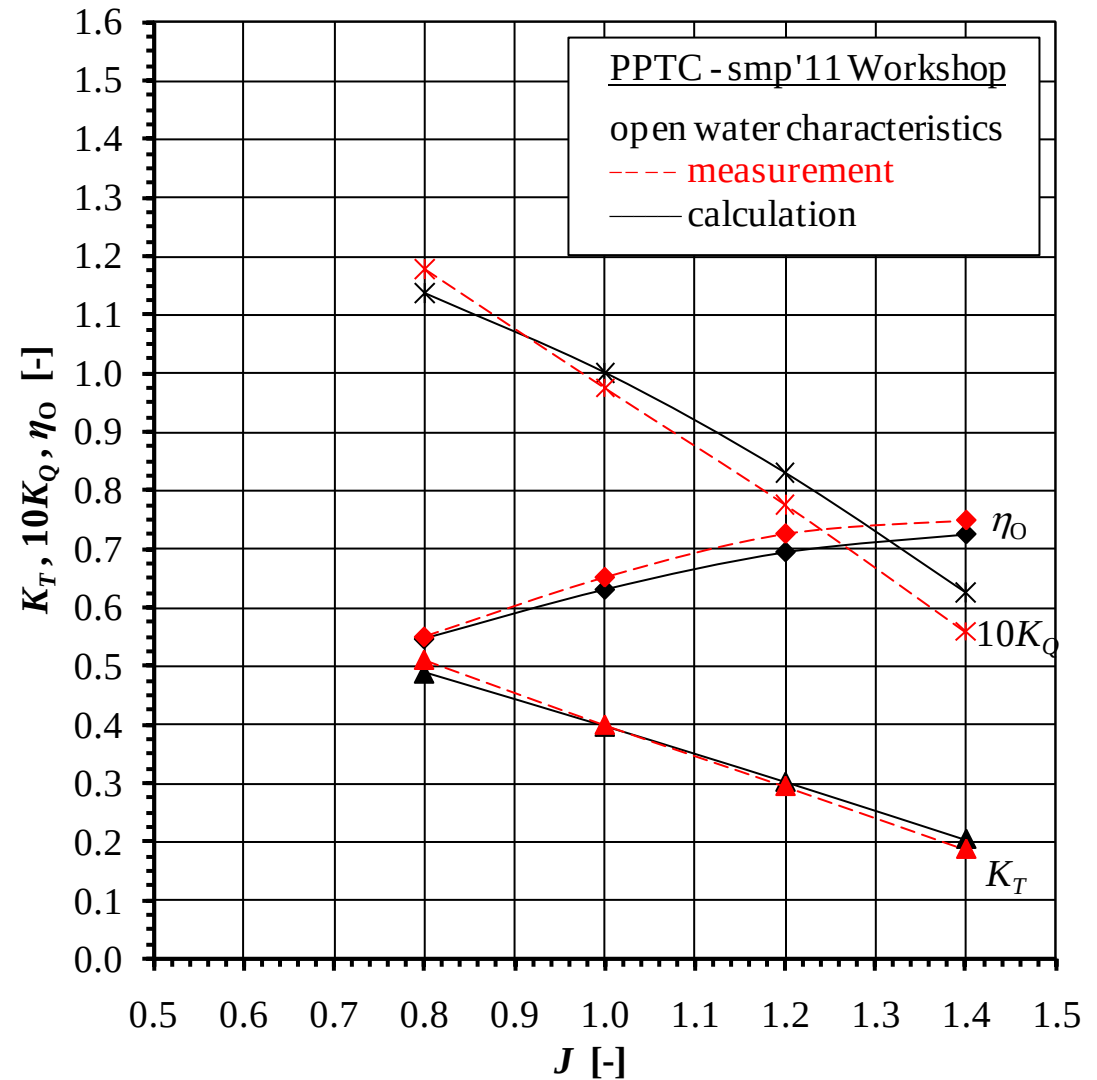
7.4 HSVA-PPB

Characteristic of propeller (model scale)

J	K_T EFD	K_T CFD	ΔK_T	ΔK_T [%]
0.60	0.6288			
0.80	0.5100	0.4888	-0.0212	-4.15
1.00	0.3994	0.3972	-0.0022	-0.54
1.20	0.2949	0.3023	0.0074	2.50
1.40	0.1878	0.2039	0.0161	8.59

J	$10K_Q$ EFD	$10K_Q$ CFD	$\Delta 10K_Q$	$\Delta 10K_Q$ [%]
0.60	1.3964			
0.80	1.1780	1.1369	-0.0411	-3.49
1.00	0.9749	1.0013	0.0264	2.71
1.20	0.7760	0.8305	0.0545	7.02
1.40	0.5588	0.6264	0.0676	12.09

J	η_O EFD	η_O CFD	$\Delta \eta_O$	$\Delta \eta_O$ [%]
0.60	0.4300			
0.80	0.5512	0.5472	-0.0040	-0.72
1.00	0.6520	0.6310	-0.0210	-3.22
1.20	0.7258	0.6947	-0.0311	-4.29
1.40	0.7487	0.7251	-0.0236	-3.15



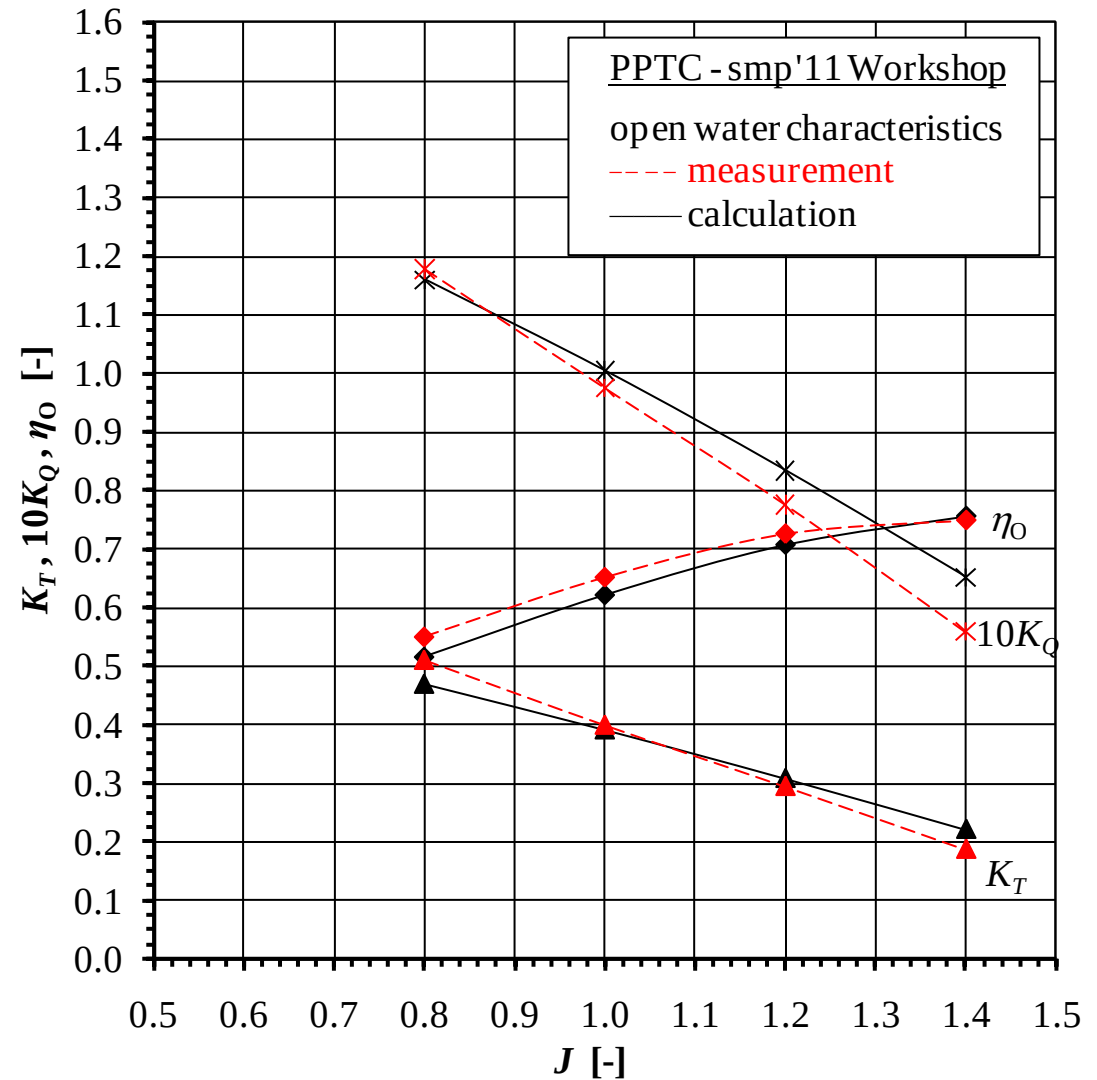
7.5 HSVA-QCM

Characteristic of propeller (model scale)

J	K_T EFD	K_T CFD	ΔK_T	ΔK_T [%]
0.60	0.6288			
0.80	0.5100	0.4700	-0.0400	-7.84
1.00	0.3994	0.3920	-0.0074	-1.85
1.20	0.2949	0.3090	0.0141	4.77
1.40	0.1878	0.2220	0.0342	18.23

J	$10K_Q$ EFD	$10K_Q$ CFD	$\Delta 10K_Q$	$\Delta 10K_Q$ [%]
0.60	1.3964			
0.80	1.1780	1.1600	-0.0180	-1.53
1.00	0.9749	1.0040	0.0291	2.99
1.20	0.7760	0.8340	0.0580	7.47
1.40	0.5588	0.6520	0.0932	16.67

J	η_O EFD	η_O CFD	$\Delta \eta_O$	$\Delta \eta_O$ [%]
0.60	0.4300			
0.80	0.5512	0.5160	-0.0352	-6.38
1.00	0.6520	0.6220	-0.0300	-4.60
1.20	0.7258	0.7080	-0.0178	-2.45
1.40	0.7487	0.7570	0.0083	1.11



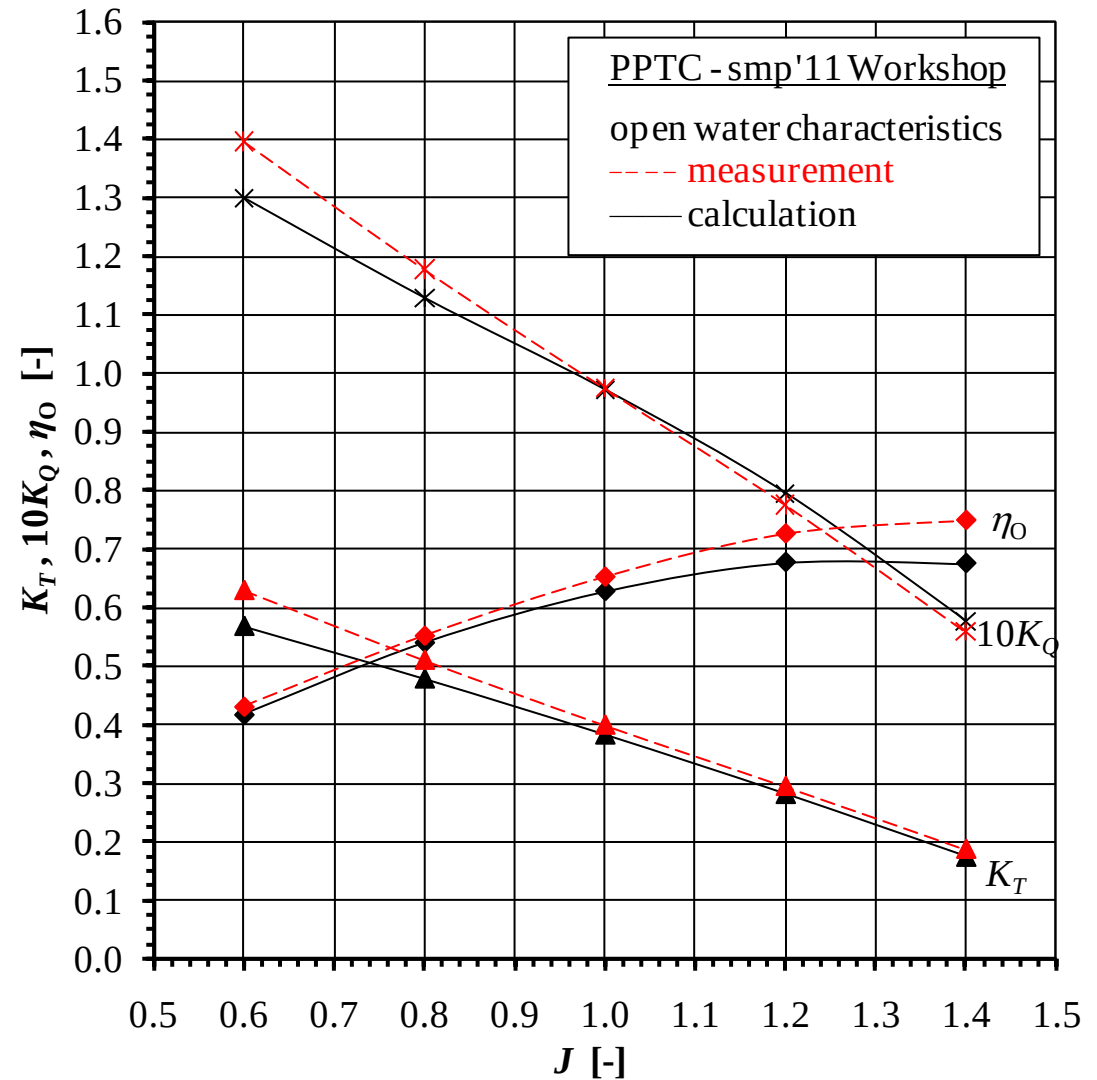
7.6 INSEAN-PFC

Characteristic of propeller (model scale)

J	K_T EFD	K_T CFD	ΔK_T	ΔK_T [%]
0.60	0.6288	0.5690	-0.0599	-9.52
0.80	0.5100	0.4791	-0.0308	-6.04
1.00	0.3994	0.3835	-0.0159	-3.97
1.20	0.2949	0.2821	-0.0128	-4.35
1.40	0.1878	0.1748	-0.0129	-6.89

J	$10K_Q$ EFD	$10K_Q$ CFD	$\Delta 10K_Q$	$\Delta 10K_Q$ [%]
0.60	1.3964	1.2994	-0.0970	-6.94
0.80	1.1780	1.1286	-0.0494	-4.20
1.00	0.9749	0.9718	-0.0030	-0.31
1.20	0.7760	0.7954	0.0194	2.50
1.40	0.5588	0.5767	0.0179	3.20

J	η_O EFD	η_O CFD	$\Delta \eta_O$	$\Delta \eta_O$ [%]
0.60	0.4300	0.4181	-0.0119	-2.77
0.80	0.5512	0.5406	-0.0106	-1.93
1.00	0.6520	0.6281	-0.0239	-3.67
1.20	0.7258	0.6773	-0.0485	-6.68
1.40	0.7487	0.6755	-0.0731	-9.77



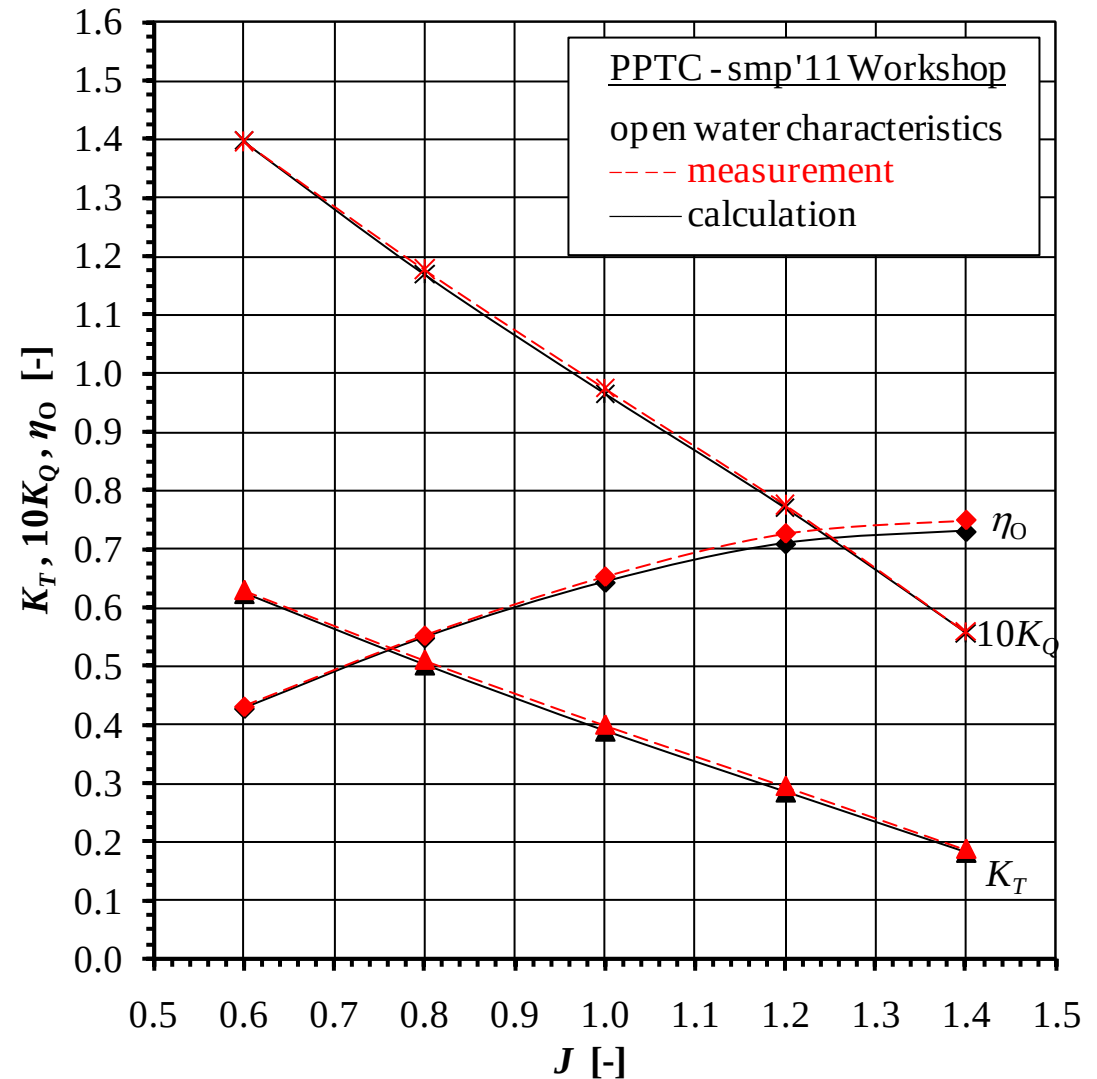
7.7 MARIC-Fluent

Characteristic of propeller (model scale)

J	K_T EFD	K_T CFD	ΔK_T	ΔK_T [%]
0.60	0.6288	0.6250	-0.0038	-0.61
0.80	0.5100	0.5030	-0.0070	-1.37
1.00	0.3994	0.3900	-0.0094	-2.35
1.20	0.2949	0.2860	-0.0089	-3.02
1.40	0.1878	0.1830	-0.0048	-2.54

J	$10K_Q$ EFD	$10K_Q$ CFD	$\Delta 10K_Q$	$\Delta 10K_Q$ [%]
0.60	1.3964	1.3970	0.0006	0.04
0.80	1.1780	1.1700	-0.0080	-0.68
1.00	0.9749	0.9660	-0.0089	-0.91
1.20	0.7760	0.7700	-0.0060	-0.78
1.40	0.5588	0.5570	-0.0018	-0.33

J	η_O EFD	η_O CFD	$\Delta \eta_O$	$\Delta \eta_O$ [%]
0.60	0.4300	0.4270	-0.0030	-0.71
0.80	0.5512	0.5480	-0.0032	-0.58
1.00	0.6520	0.6430	-0.0090	-1.40
1.20	0.7258	0.7090	-0.0168	-2.37
1.40	0.7487	0.7300	-0.0187	-2.56



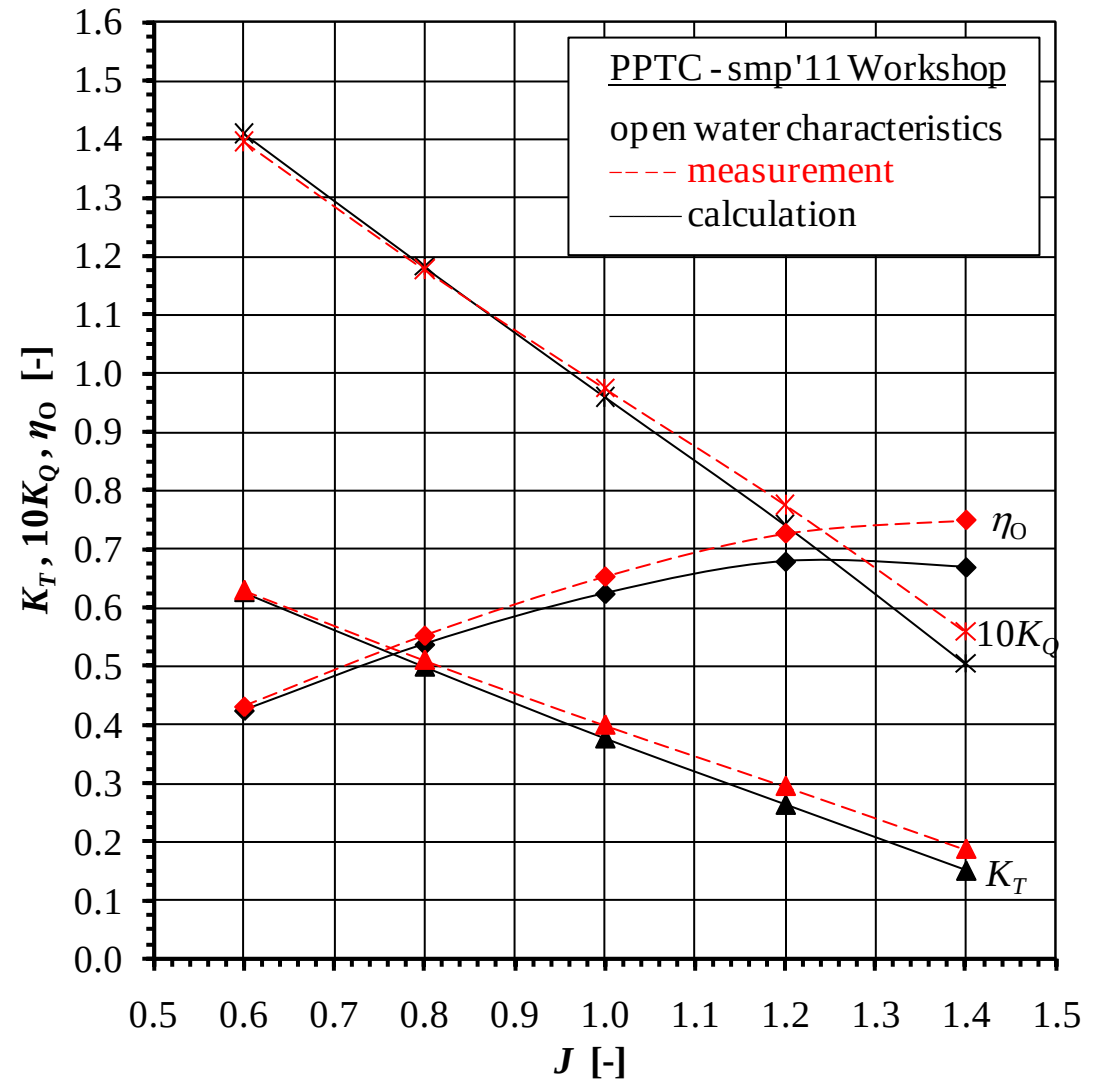
7.8 SSPA-Fluent

Characteristic of propeller (model scale)

J	K_T EFD	K_T CFD	ΔK_T	ΔK_T [%]
0.60	0.6288	0.6258	-0.0030	-0.48
0.80	0.5100	0.4990	-0.0110	-2.15
1.00	0.3994	0.3769	-0.0225	-5.63
1.20	0.2949	0.2640	-0.0309	-10.48
1.40	0.1878	0.1518	-0.0360	-19.15

J	$10K_Q$ EFD	$10K_Q$ C	$\Delta 10K_Q$	$\Delta 10K_Q$ [%]
0.60	1.3964	1.4096	0.0132	0.95
0.80	1.1780	1.1822	0.0041	0.35
1.00	0.9749	0.9605	-0.0143	-1.47
1.20	0.7760	0.7418	-0.0343	-4.42
1.40	0.5588	0.5049	-0.0540	-9.66

J	η_O EFD	η_O CFD	$\Delta \eta_O$	$\Delta \eta_O$ [%]
0.60	0.4300	0.4240	-0.0061	-1.41
0.80	0.5512	0.5375	-0.0137	-2.49
1.00	0.6520	0.6245	-0.0275	-4.22
1.20	0.7258	0.6797	-0.0461	-6.35
1.40	0.7487	0.6700	-0.0787	-10.51



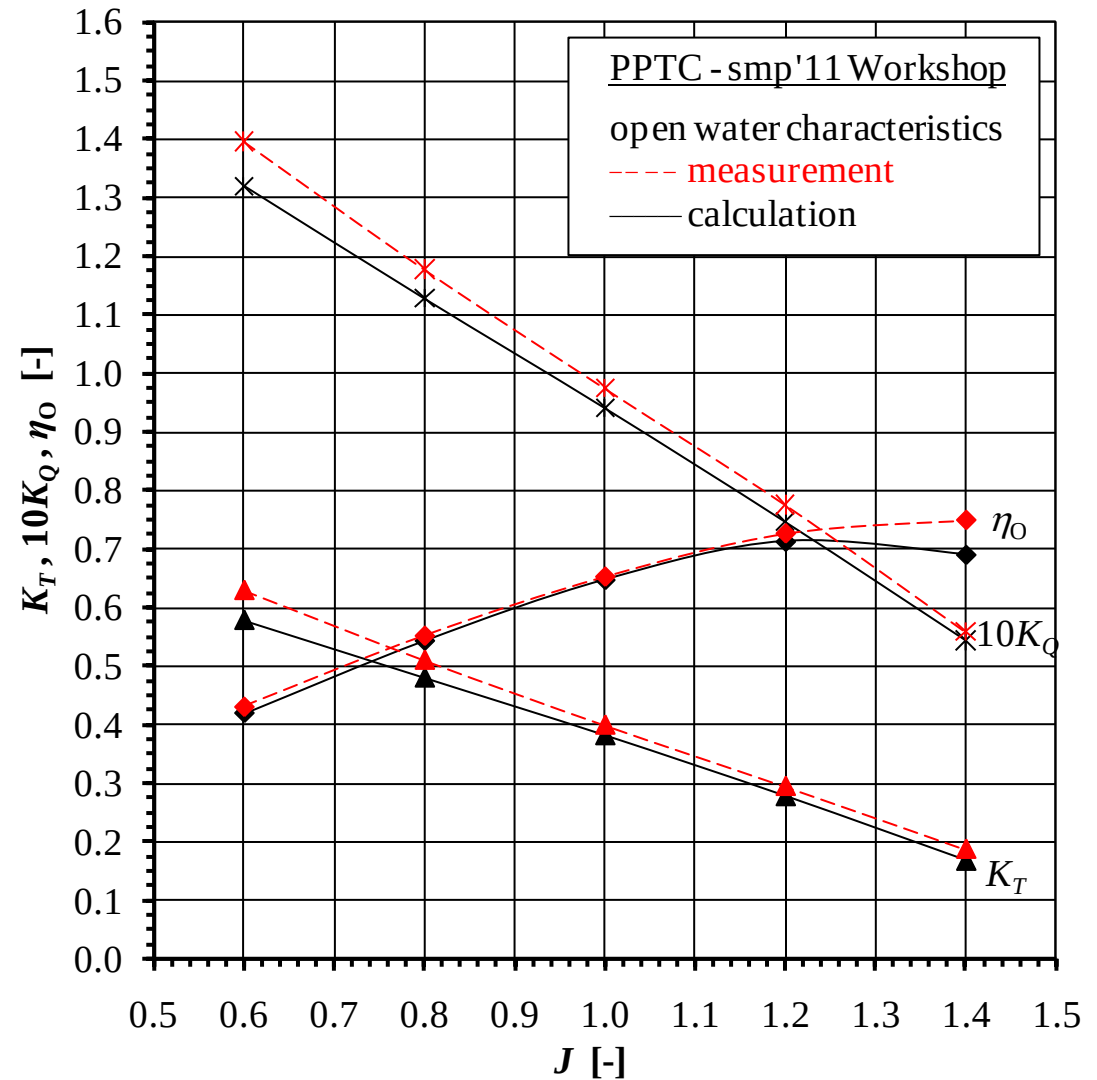
7.9 SVA-Vortex

Characteristic of propeller (model scale)

J	K_T EFD	K_T CFD	ΔK_T	ΔK_T [%]
0.60	0.6288	0.5796	-0.0492	-7.82
0.80	0.5100	0.4814	-0.0285	-5.60
1.00	0.3994	0.3831	-0.0163	-4.08
1.20	0.2949	0.2791	-0.0159	-5.38
1.40	0.1878	0.1689	-0.0189	-10.07

J	$10K_Q$ EFD	$10K_Q$ CFD	$\Delta 10K_Q$	$\Delta 10K_Q$ [%]
0.60	1.3964	1.3203	-0.0761	-5.45
0.80	1.1780	1.1278	-0.0502	-4.26
1.00	0.9749	0.9407	-0.0342	-3.51
1.20	0.7760	0.7465	-0.0296	-3.81
1.40	0.5588	0.5442	-0.0146	-2.61

J	η_O EFD	η_O CFD	$\Delta \eta_O$	$\Delta \eta_O$ [%]
0.60	0.4300	0.4192	-0.0108	-2.51
0.80	0.5512	0.5435	-0.0077	-1.39
1.00	0.6520	0.6481	-0.0039	-0.59
1.20	0.7258	0.7140	-0.0119	-1.63
1.40	0.7487	0.6914	-0.0573	-7.65



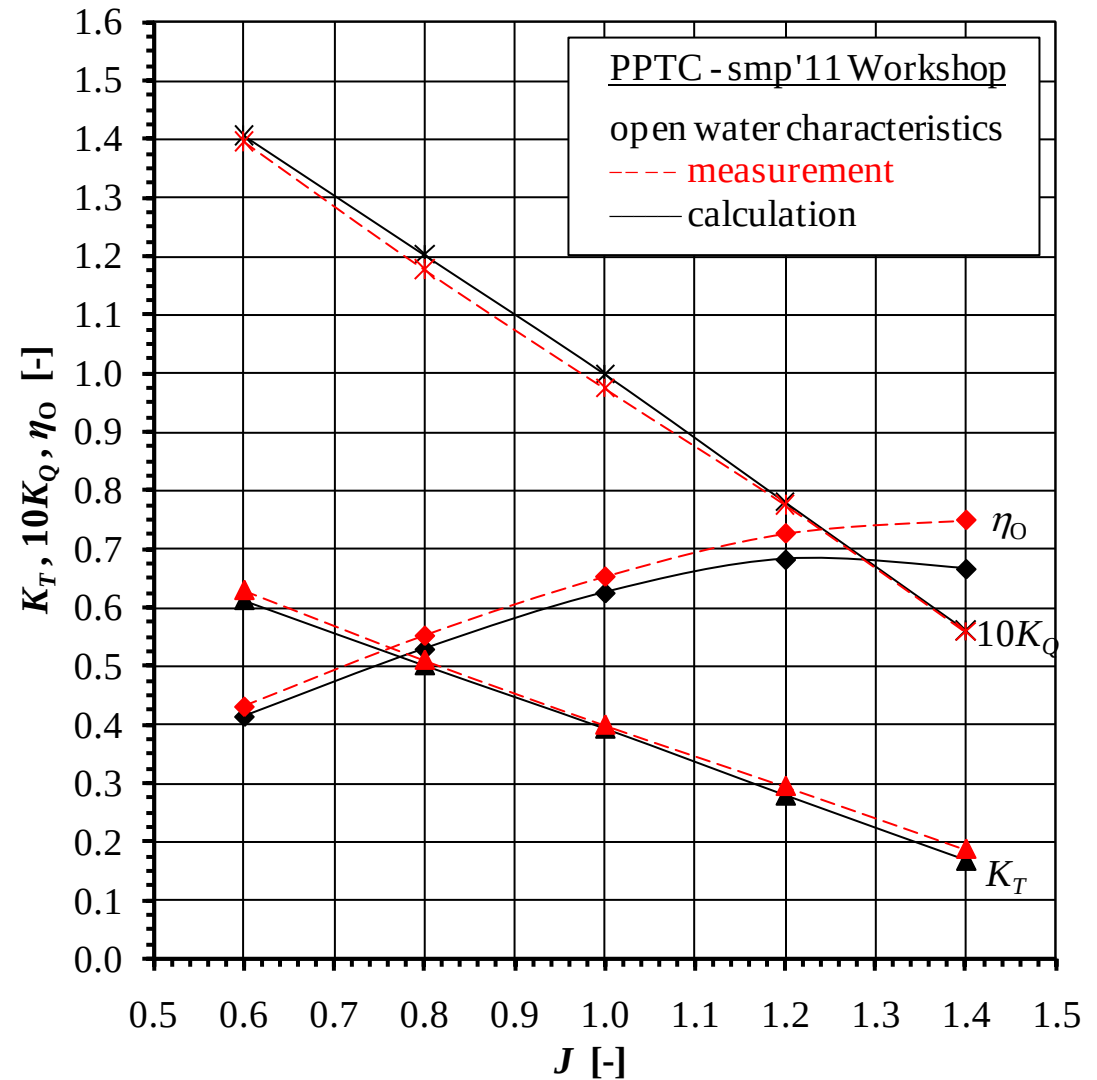
7.10 TUHH-FreSCO

Characteristic of propeller (model scale)

J	K_T EFD	K_T CFD	ΔK_T	ΔK_T [%]
0.60	0.629	0.612	-0.017	-2.676
0.80	0.510	0.501	-0.009	-1.758
1.00	0.399	0.393	-0.006	-1.595
1.20	0.295	0.279	-0.016	-5.398
1.40	0.188	0.168	-0.020	-10.525

J	$10K_Q$ EFD	$10K_Q$ CFD	$\Delta 10K_Q$	$\Delta 10K_Q$ [%]
0.60	1.396	1.407	0.011	0.761
0.80	1.178	1.203	0.025	2.122
1.00	0.975	0.999	0.024	2.474
1.20	0.776	0.780	0.004	0.509
1.40	0.559	0.561	0.002	0.390

J	η_O EFD	η_O CFD	$\Delta \eta_O$	$\Delta \eta_O$ [%]
0.60	0.430	0.415	-0.015	-3.496
0.80	0.551	0.530	-0.021	-3.845
1.00	0.652	0.626	-0.026	-3.987
1.20	0.726	0.683	-0.043	-5.897
1.40	0.749	0.667	-0.082	-10.907



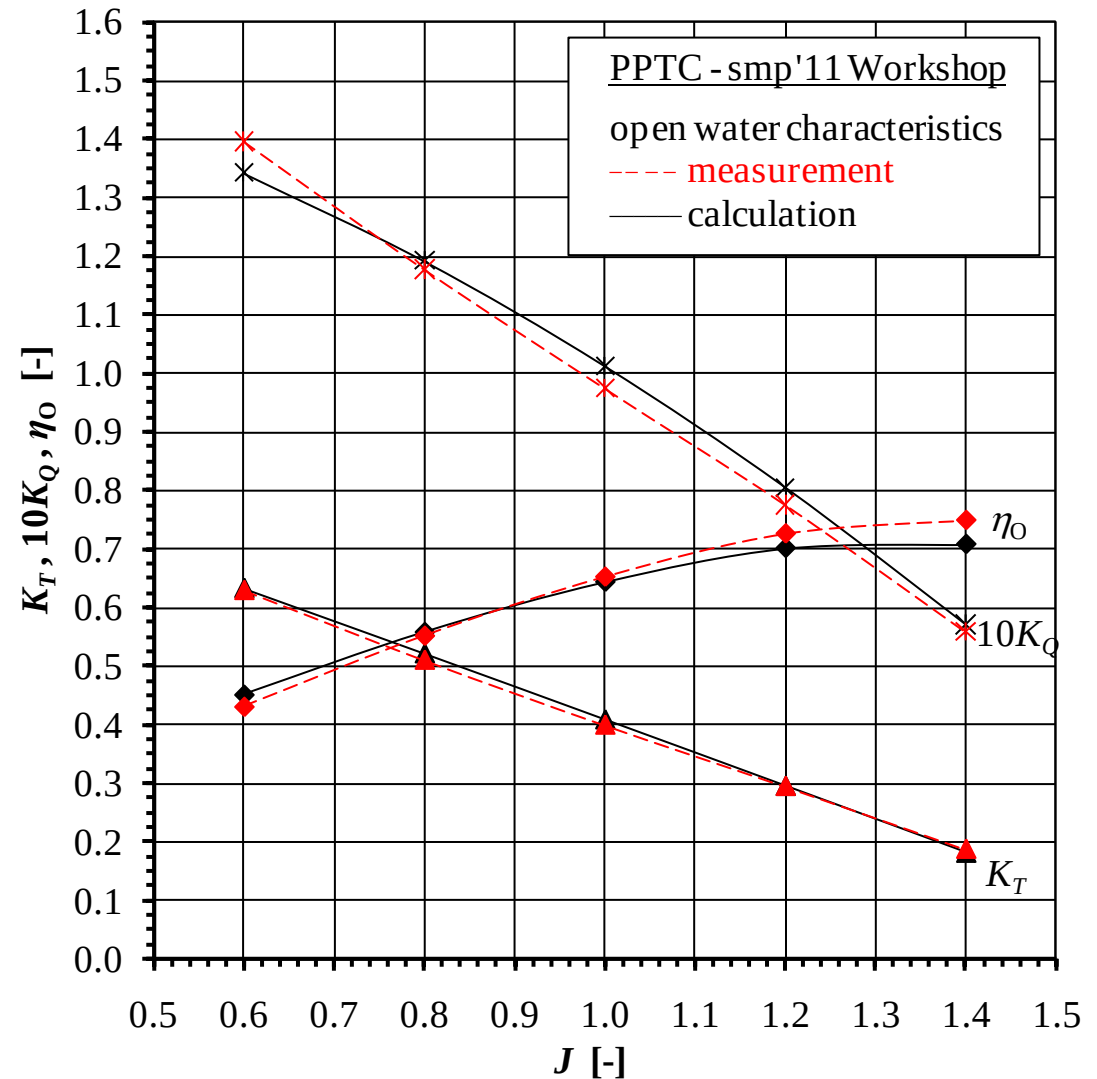
7.11 UniGenua-Panel

Characteristic of propeller (model scale)

J	K_T EFD	K_T CFD	ΔK_T	ΔK_T [%]
0.60	0.6288	0.6343	0.0054	0.86
0.80	0.5100	0.5225	0.0126	2.47
1.00	0.3994	0.4096	0.0102	2.56
1.20	0.2949	0.2959	0.0009	0.32
1.40	0.1878	0.1816	-0.0062	-3.30

J	$10K_Q$ EFD	$10K_Q$ CFD	$\Delta 10K_Q$	$\Delta 10K_Q$ [%]
0.60	1.3964	1.3426	-0.0538	-3.85
0.80	1.1780	1.1924	0.0144	1.22
1.00	0.9749	1.0131	0.0382	3.92
1.20	0.7760	0.8060	0.0299	3.85
1.40	0.5588	0.5716	0.0128	2.29

J	η_O EFD	η_O CFD	$\Delta \eta_O$	$\Delta \eta_O$ [%]
0.60	0.4300	0.4511	0.0211	4.91
0.80	0.5512	0.5580	0.0068	1.23
1.00	0.6520	0.6435	-0.0085	-1.31
1.20	0.7258	0.7011	-0.0247	-3.41
1.40	0.7487	0.7077	-0.0409	-5.46



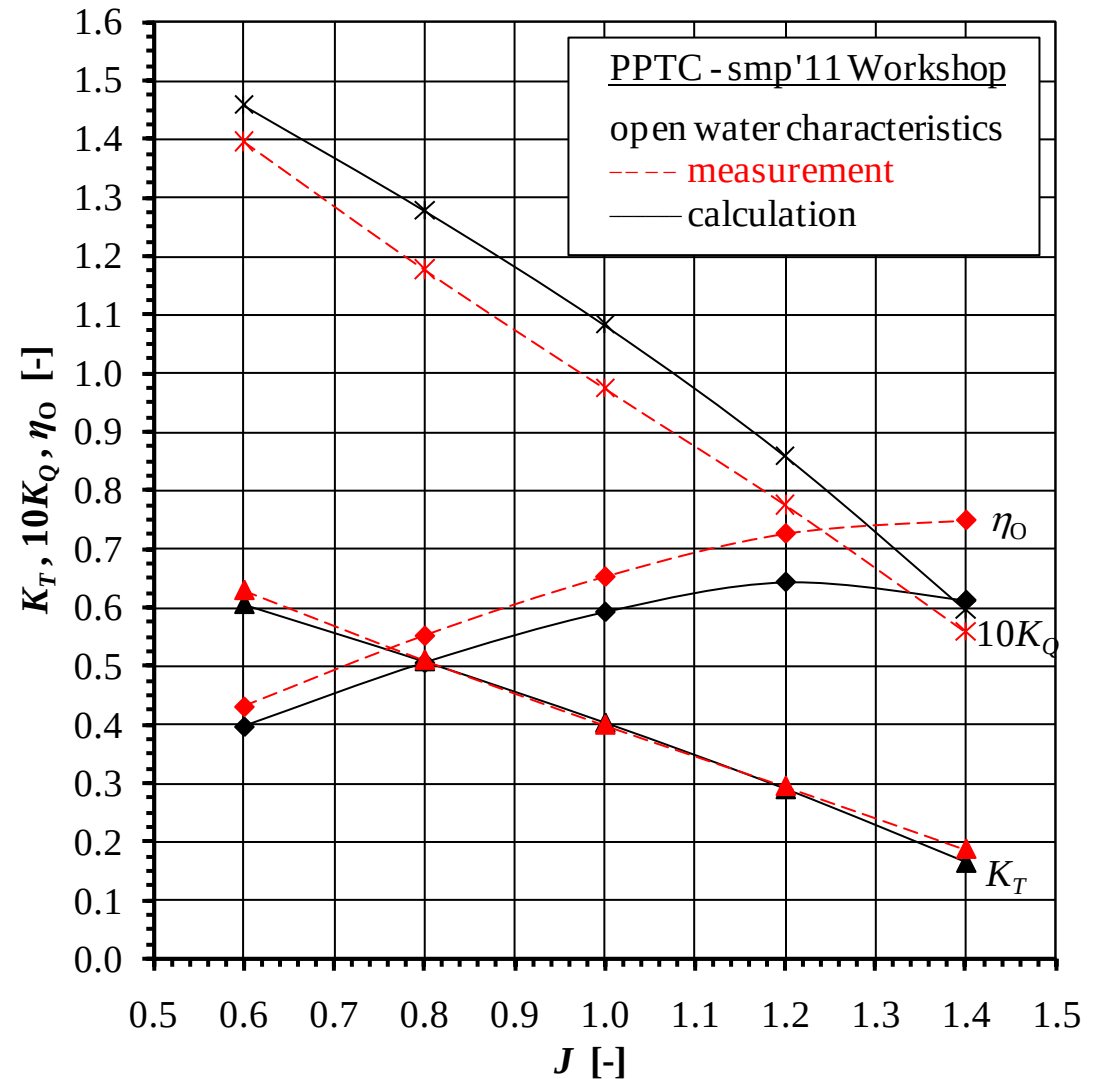
7.12 UniGenua-OpenFOAM

Characteristic of propeller (model scale)

J	K_T EFD	K_T CFD	ΔK_T	ΔK_T [%]
0.60	0.6288	0.6067	-0.0222	-3.53
0.80	0.5100	0.5088	-0.0012	-0.23
1.00	0.3994	0.4039	0.0045	1.13
1.20	0.2949	0.2902	-0.0048	-1.61
1.40	0.1878	0.1643	-0.0234	-12.47

J	$10K_Q$ EFD	$10K_Q$ CFD	$\Delta 10K_Q$	$\Delta 10K_Q$ [%]
0.60	1.3964	1.4592	0.0628	4.50
0.80	1.1780	1.2796	0.1016	8.63
1.00	0.9749	1.0837	0.1088	11.16
1.20	0.7760	0.8599	0.0839	10.81
1.40	0.5588	0.5970	0.0381	6.83

J	η_O EFD	η_O CFD	$\Delta \eta_O$	$\Delta \eta_O$ [%]
0.60	0.4300	0.3970	-0.0330	-7.68
0.80	0.5512	0.5063	-0.0449	-8.15
1.00	0.6520	0.5932	-0.0588	-9.03
1.20	0.7258	0.6445	-0.0813	-11.21
1.40	0.7487	0.6134	-0.1353	-18.07



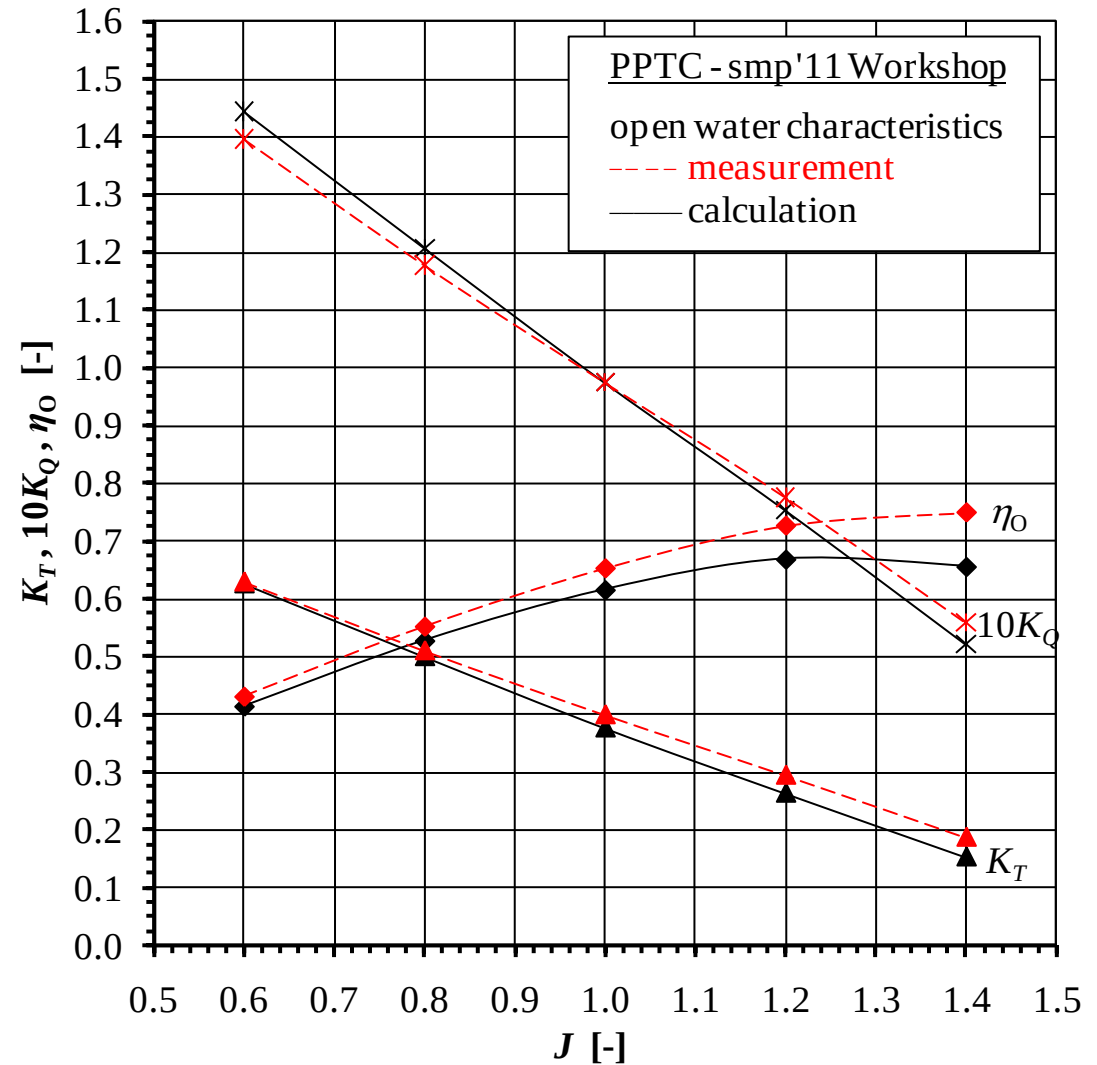
7.13 UniGenua-StarCCM(kw)

Characteristic of propeller (model scale)

J	K_T EFD	K_T CFD	ΔK_T	ΔK_T [%]
0.60	0.6288	0.6269	-0.0020	-0.31
0.80	0.5100	0.5007	-0.0093	-1.82
1.00	0.3994	0.3770	-0.0223	-5.60
1.20	0.2949	0.2639	-0.0310	-10.52
1.40	0.1878	0.1534	-0.0343	-18.29

J	$10K_Q$ EFD	$10K_Q$ CFD	$\Delta 10K_Q$	$\Delta 10K_Q$ [%]
0.60	1.3964	1.4443	0.0479	3.43
0.80	1.1780	1.2076	0.0296	2.51
1.00	0.9749	0.9743	-0.0005	-0.06
1.20	0.7760	0.7535	-0.0225	-2.90
1.40	0.5588	0.5211	-0.0378	-6.76

J	η_o EFD	η_o CFD	$\Delta \eta_o$	$\Delta \eta_o$ [%]
0.60	0.4300	0.4145	-0.0156	-3.62
0.80	0.5512	0.5279	-0.0233	-4.22
1.00	0.6520	0.6159	-0.0361	-5.54
1.20	0.7258	0.6689	-0.0569	-7.84
1.40	0.7487	0.6561	-0.0926	-12.37



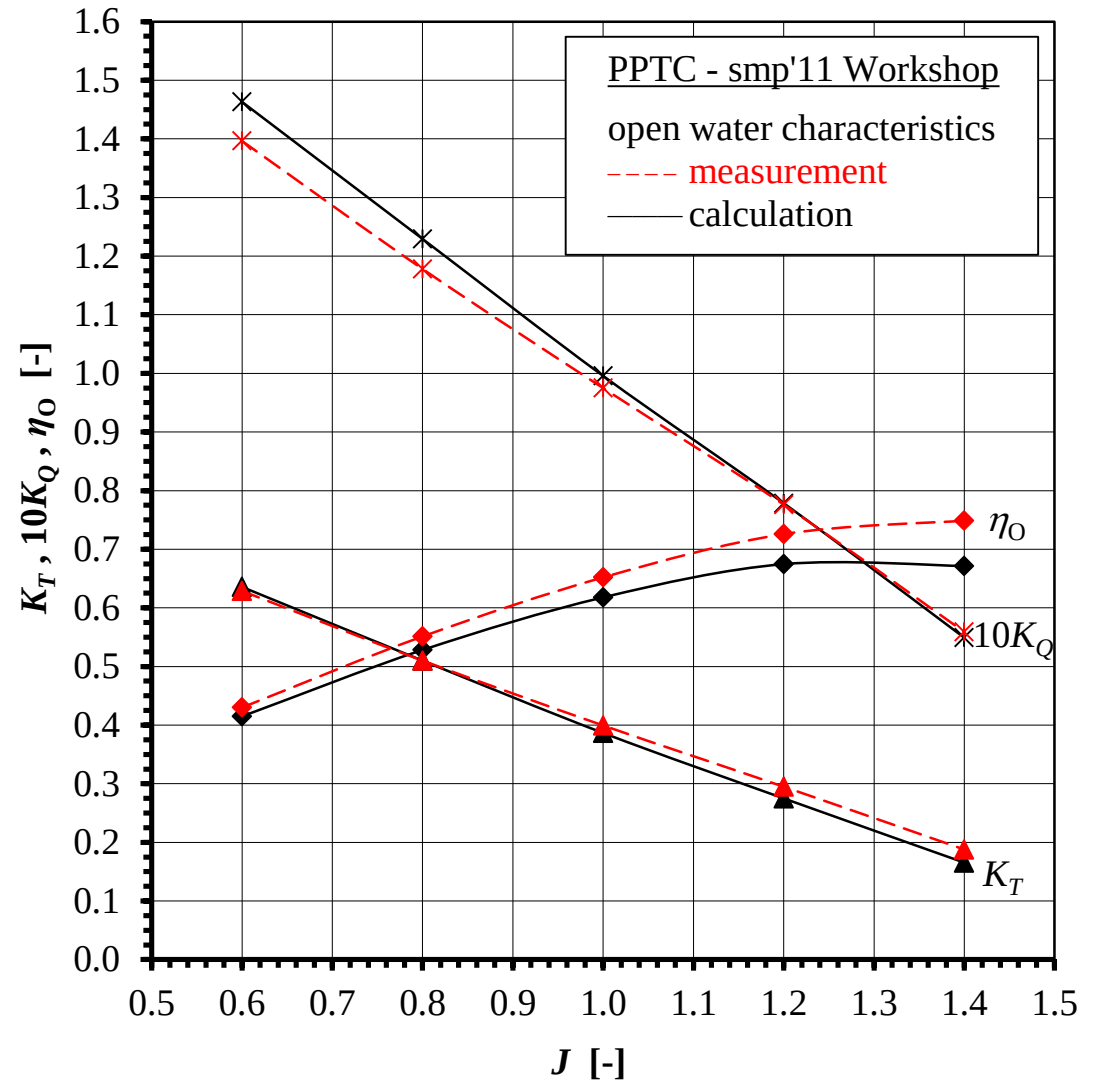
7.14 UniGenua-StarCCM(ke)

Characteristic of propeller (model scale)

J	K_T EFD	K_T CFD	ΔK_T	ΔK_T [%]
0.60	0.6288	0.6357	0.0069	1.09
0.80	0.5100	0.5099	0.0000	-0.01
1.00	0.3994	0.3866	-0.0128	-3.21
1.20	0.2949	0.2750	-0.0200	-6.77
1.40	0.1878	0.1654	-0.0223	-11.90

J	$10K_Q$ EFD	$10K_Q$ CFD	$\Delta 10K_Q$	$\Delta 10K_Q$ [%]
0.60	1.3964	1.4630	0.0667	4.77
0.80	1.1780	1.2292	0.0512	4.35
1.00	0.9749	0.9956	0.0208	2.13
1.20	0.7760	0.7785	0.0024	0.31
1.40	0.5588	0.5491	-0.0098	-1.75

J	η_o EFD	η_o CFD	$\Delta \eta_o$	$\Delta \eta_o$ [%]
0.60	0.4300	0.4149	-0.0151	-3.52
0.80	0.5512	0.5282	-0.0230	-4.18
1.00	0.6520	0.6179	-0.0341	-5.23
1.20	0.7258	0.6746	-0.0512	-7.05
1.40	0.7487	0.6713	-0.0773	-10.33



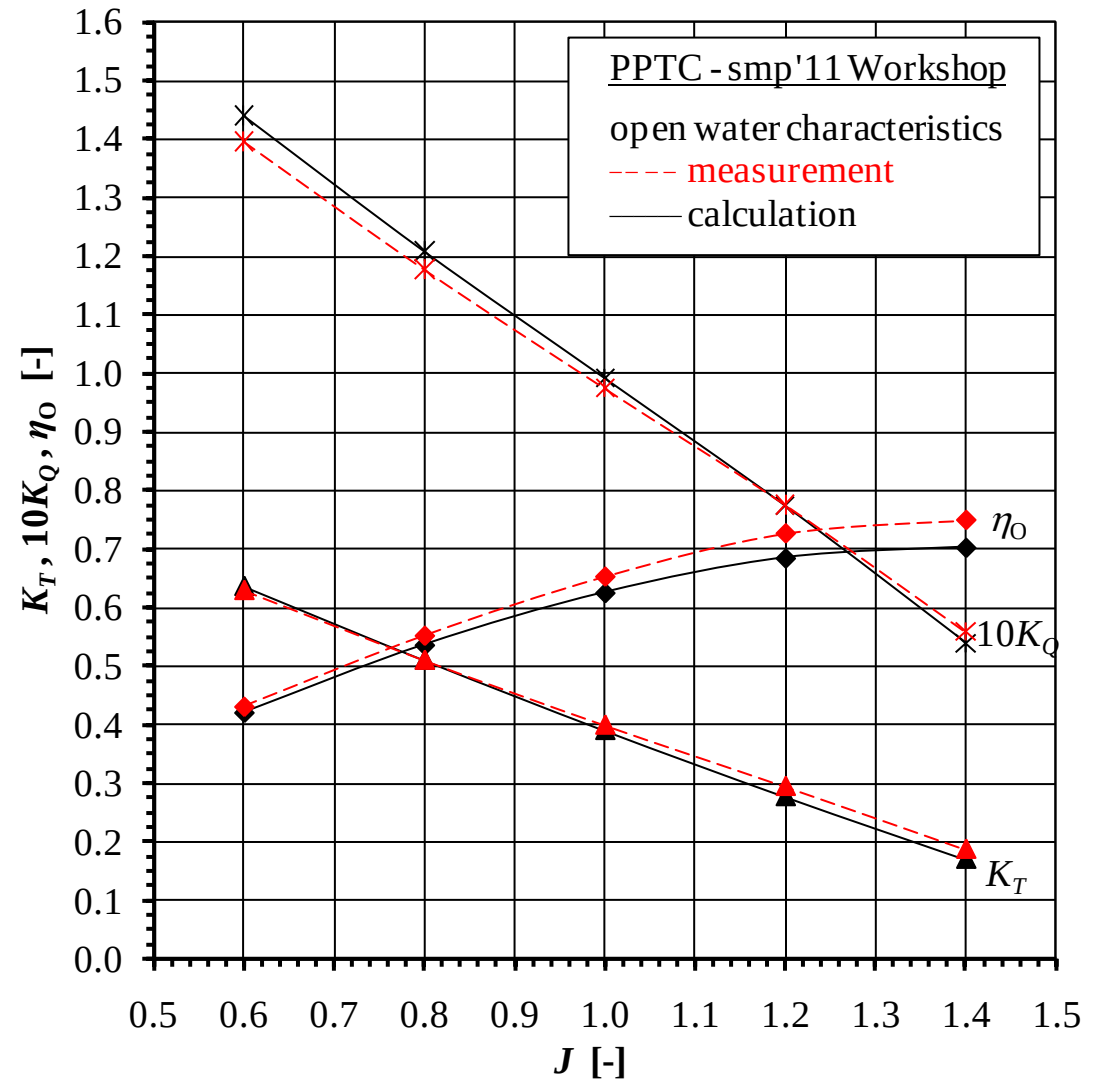
7.15 UniTriest-CFX

Characteristic of propeller (model scale)

J	K_T EFD	K_T CFD	ΔK_T	ΔK_T [%]
0.60	0.6288	0.6360	0.0072	1.14
0.80	0.5100	0.5100	0.0000	0.01
1.00	0.3994	0.3900	-0.0094	-2.35
1.20	0.2949	0.2770	-0.0179	-6.08
1.40	0.1878	0.1700	-0.0178	-9.46

J	$10K_Q$ EFD	$10K_Q$ CFD	$\Delta 10K_Q$	$\Delta 10K_Q$ [%]
0.60	1.3964	1.4410	0.0446	3.20
0.80	1.1780	1.2090	0.0310	2.63
1.00	0.9749	0.9920	0.0171	1.76
1.20	0.7760	0.7740	-0.0020	-0.26
1.40	0.5588	0.5400	-0.0188	-3.37

J	η_O EFD	η_O CFD	$\Delta \eta_O$	$\Delta \eta_O$ [%]
0.60	0.4300	0.4220	-0.0080	-1.87
0.80	0.5512	0.5370	-0.0142	-2.57
1.00	0.6520	0.6260	-0.0260	-3.99
1.20	0.7258	0.6850	-0.0408	-5.62
1.40	0.7487	0.7030	-0.0457	-6.10



7.16 VicusDT-StarCCM

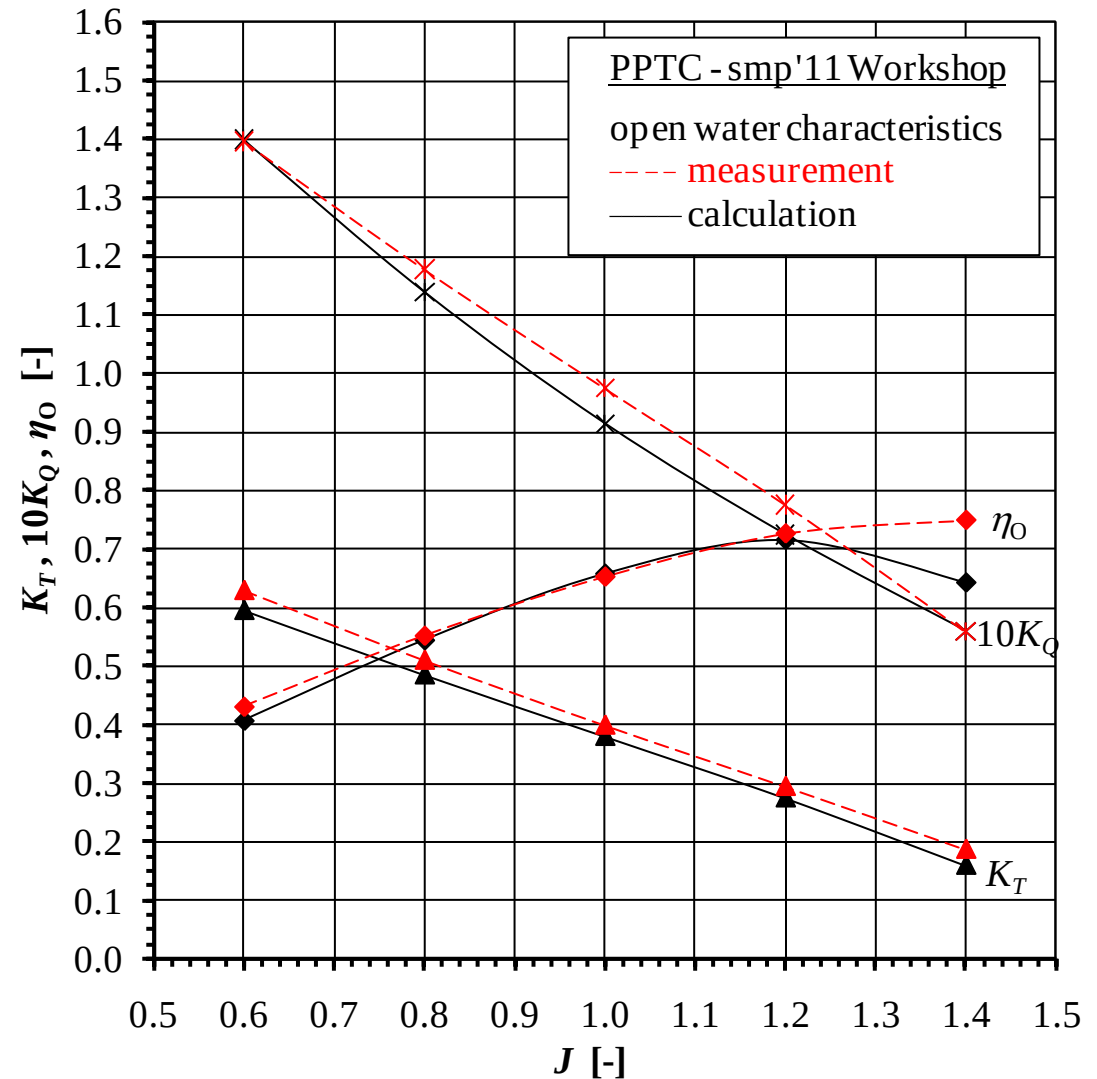
Characteristic of propeller (model scale)

J	K_T EFD	K_T CFD	ΔK_T	ΔK_T [%]
0.60	0.6288	0.5950	-0.0338	-5.38
0.80	0.5100	0.4850	-0.0250	-4.90
1.00	0.3994	0.3800	-0.0194	-4.85
1.20	0.2949	0.2750	-0.0199	-6.75
1.40	0.1878	0.1600	-0.0278	-14.79

J	$10K_Q$ EFD	$10K_Q$ CFD	$\Delta 10K_Q$	$\Delta 10K_Q$ [%]
0.60	1.3964	1.4000	0.0036	0.26
0.80	1.1780	1.1400	-0.0380	-3.23
1.00	0.9749	0.9150	-0.0599	-6.14
1.20	0.7760	0.7250	-0.0510	-6.58
1.40	0.5588	0.5600	0.0012	0.21

J	η_O EFD	η_O CFD	$\Delta \eta_O$	$\Delta \eta_O$ [%]
0.60	0.4300	0.4070	-0.0230	-5.36
0.80	0.5512	0.5450	-0.0062	-1.12
1.00	0.6520	0.6590	0.0070	1.07
1.20	0.7258	0.7170	-0.0088	-1.21
1.40	0.7487	0.6440	-0.1047	-13.98

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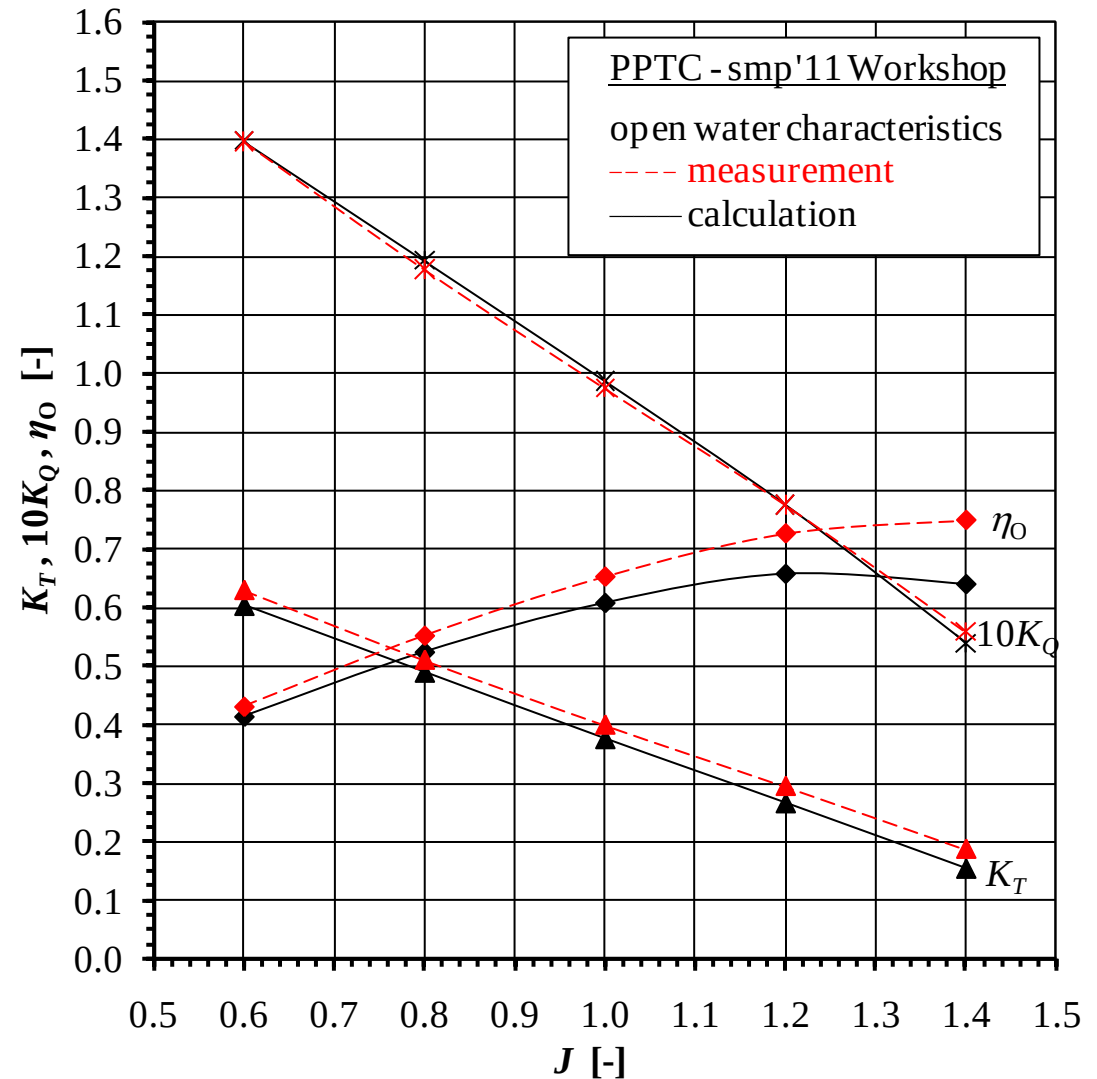
7.17 VOITH-Comet

Characteristic of propeller (model scale)

J	K_T EFD	K_T CFD	ΔK_T	ΔK_T [%]
0.60	0.6288	0.6044	-0.0244	-3.88
0.80	0.5100	0.4901	-0.0198	-3.89
1.00	0.3994	0.3765	-0.0229	-5.74
1.20	0.2949	0.2667	-0.0283	-9.58
1.40	0.1878	0.1547	-0.0331	-17.62

J	$10K_Q$ EFD	$10K_Q$ CFD	$\Delta 10K_Q$	$\Delta 10K_Q$ [%]
0.60	1.3964	1.3977	0.0013	0.10
0.80	1.1780	1.1932	0.0152	1.29
1.00	0.9749	0.9874	0.0125	1.28
1.20	0.7760	0.7757	-0.0003	-0.04
1.40	0.5588	0.5394	-0.0194	-3.48

J	η_O EFD	η_O CFD	$\Delta \eta_O$	$\Delta \eta_O$ [%]
0.60	0.4300	0.4130	-0.0171	-3.97
0.80	0.5512	0.5230	-0.0282	-5.11
1.00	0.6520	0.6068	-0.0452	-6.93
1.20	0.7258	0.6565	-0.0693	-9.55
1.40	0.7487	0.6390	-0.1097	-14.65



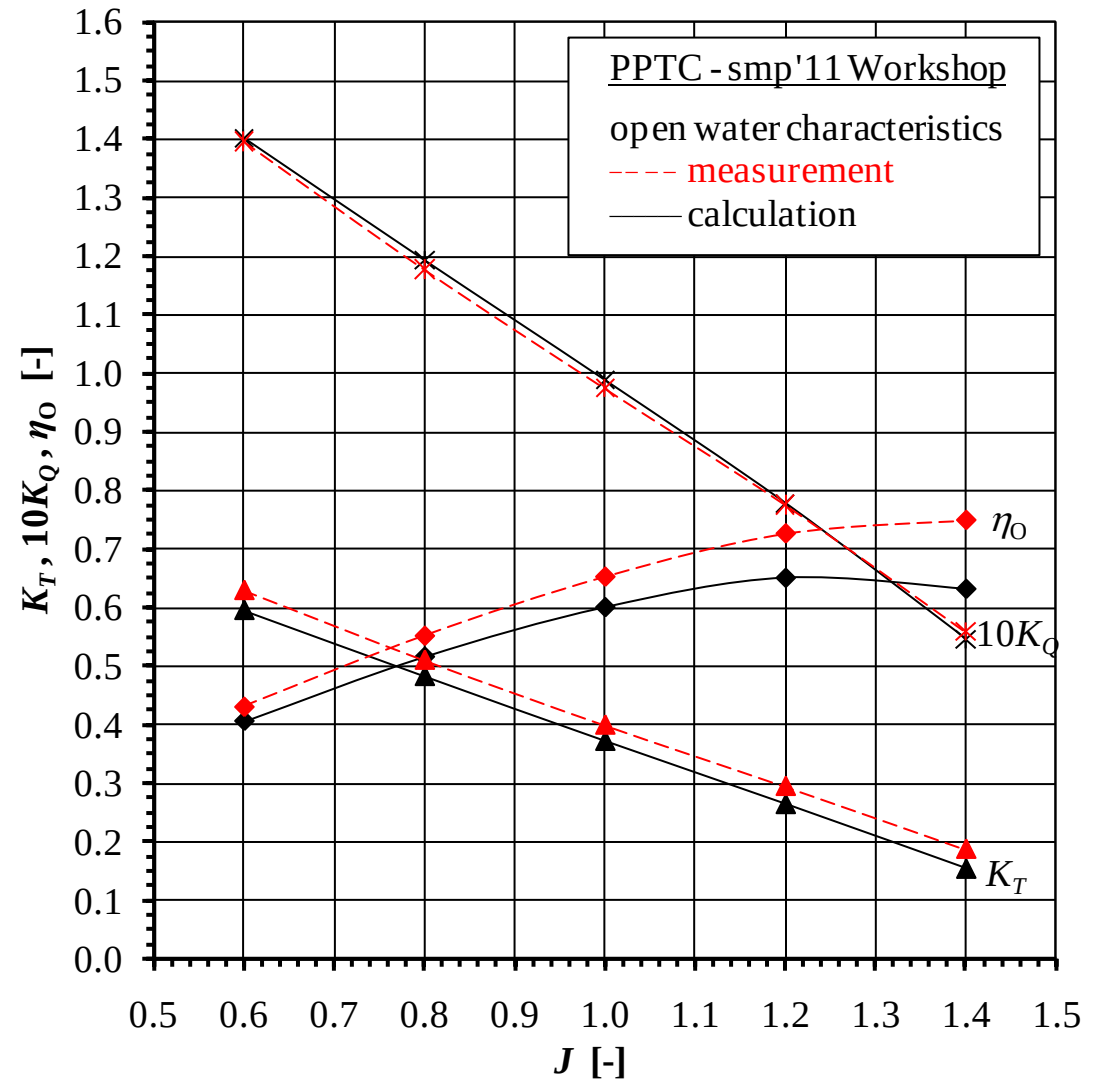
7.18 VOITH-OpenFOAM

Characteristic of propeller (model scale)

J	K_T EFD	K_T CFD	ΔK_T	ΔK_T [%]
0.60	0.6288	0.5957	-0.0331	-5.27
0.80	0.5100	0.4832	-0.0268	-5.25
1.00	0.3994	0.3727	-0.0267	-6.68
1.20	0.2949	0.2650	-0.0299	-10.15
1.40	0.1878	0.1548	-0.0330	-17.56

J	$10K_Q$ EFD	$10K_Q$ CFD	$\Delta 10K_Q$	$\Delta 10K_Q$ [%]
0.60	1.3964	1.4021	0.0057	0.41
0.80	1.1780	1.1933	0.0153	1.30
1.00	0.9749	0.9884	0.0135	1.39
1.20	0.7760	0.7783	0.0023	0.29
1.40	0.5588	0.5463	-0.0125	-2.24

J	η_O EFD	η_O CFD	$\Delta \eta_O$	$\Delta \eta_O$ [%]
0.60	0.4300	0.4057	-0.0243	-5.66
0.80	0.5512	0.5156	-0.0356	-6.46
1.00	0.6520	0.6001	-0.0519	-7.96
1.20	0.7258	0.6503	-0.0755	-10.40
1.40	0.7487	0.6314	-0.1173	-15.66



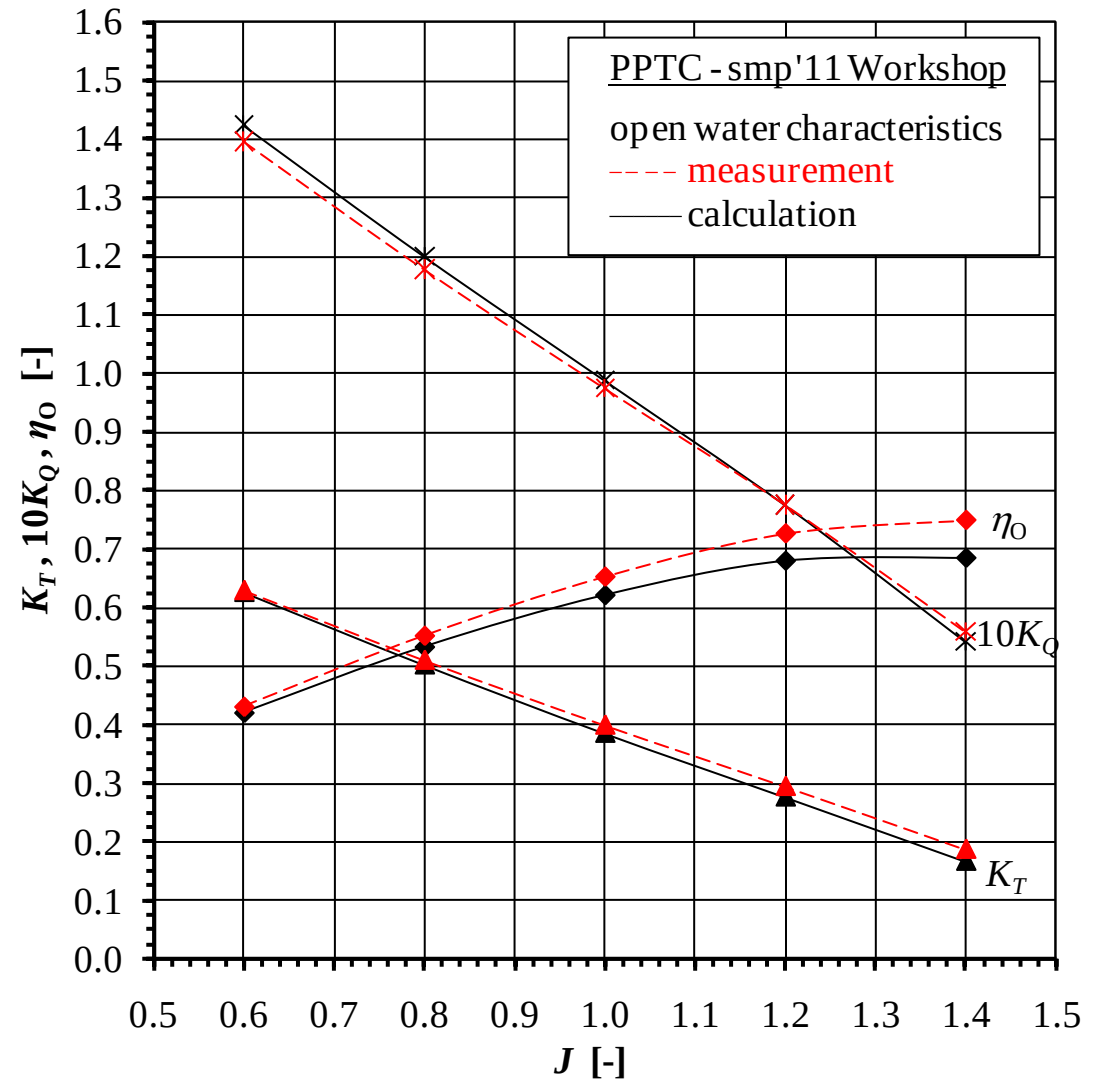
7.19 VTT-FinFlo

Characteristic of propeller (model scale)

J	K_T EFD	K_T CFD	ΔK_T	ΔK_T [%]
0.60	0.6288	0.6270	-0.0018	-0.29
0.80	0.5100	0.5030	-0.0070	-1.37
1.00	0.3994	0.3860	-0.0134	-3.35
1.20	0.2949	0.2770	-0.0179	-6.08
1.40	0.1878	0.1670	-0.0208	-11.06

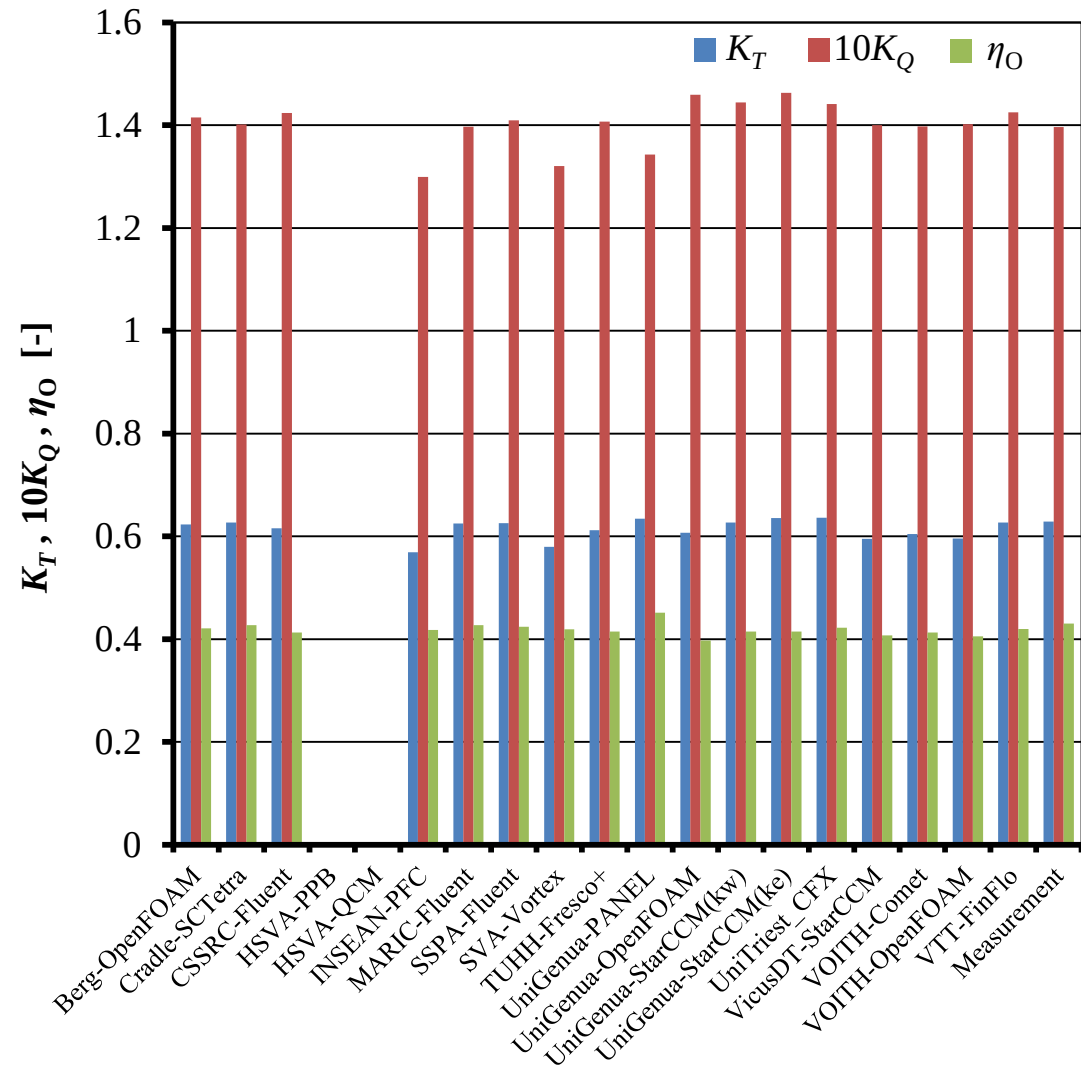
J	$10K_Q$ EFD	$10K_Q$ CFD	$\Delta 10K_Q$	$\Delta 10K_Q$ [%]
0.60	1.3964	1.4250	0.0286	2.05
0.80	1.1780	1.2010	0.0230	1.95
1.00	0.9749	0.9890	0.0141	1.45
1.20	0.7760	0.7760	0.0000	-0.01
1.40	0.5588	0.5420	-0.0168	-3.01

J	η_O EFD	η_O CFD	$\Delta \eta_O$	$\Delta \eta_O$ [%]
0.60	0.4300	0.4200	-0.0100	-2.33
0.80	0.5512	0.5330	-0.0182	-3.30
1.00	0.6520	0.6220	-0.0300	-4.60
1.20	0.7258	0.6810	-0.0448	-6.17
1.40	0.7487	0.6860	-0.0627	-8.37



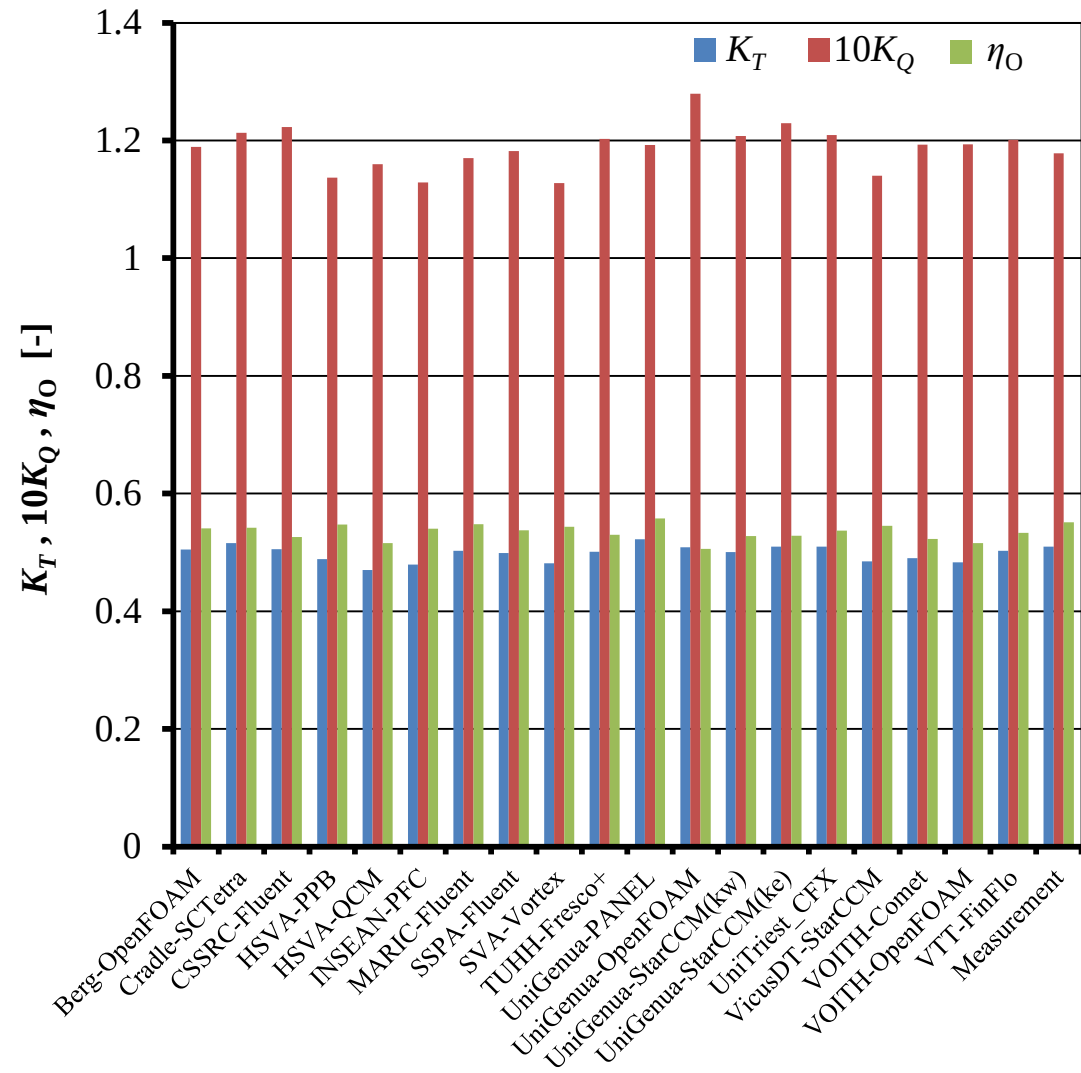
7.20 Comparison of open water characteristics, $J = 0.6$

	K_T [-]	$10K_Q$ [-]	η_o [-]
Berg-OpenFOAM	0.623	1.415	0.421
Cradle-SCTetra	0.627	1.401	0.427
CSSRC-Fluent	0.616	1.424	0.413
HSVA-PPB			
HSVA-QCM			
INSEAN-PFC	0.569	1.299	0.418
MARIC-Fluent	0.625	1.397	0.427
SSPA-Fluent	0.626	1.410	0.424
SVA-Vortex	0.580	1.320	0.419
TUHH-Fresco+	0.612	1.407	0.415
UniGenua-PANEL	0.634	1.343	0.451
UniGenua-OpenFOAM	0.607	1.459	0.397
UniGenua-StarCCM(kw)	0.627	1.444	0.414
UniGenua-StarCCM(ke)	0.636	1.463	0.415
UniTriest	0.636	1.441	0.422
VicusDT-StarCCM	0.595	1.400	0.407
VOITH-Comet	0.604	1.398	0.413
VOITH-OpenFOAM	0.596	1.402	0.406
VTT-FinFlo	0.627	1.425	0.420
Measurement	0.629	1.396	0.430



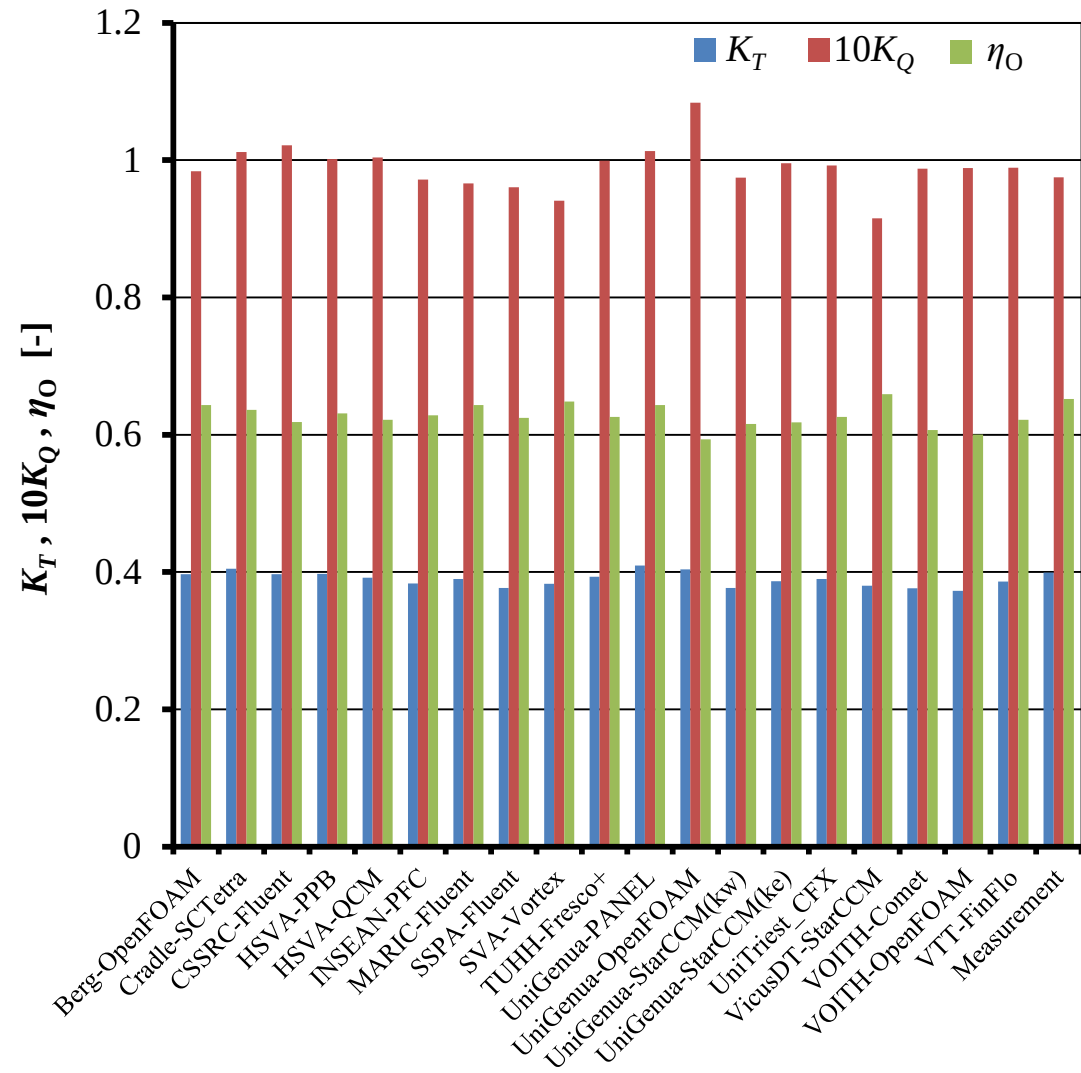
7.21 Comparison of open water characteristics, $J = 0.8$

	K_T [-]	$10K_Q$ [-]	η_o [-]
Berg-OpenFOAM	0.505	1.189	0.541
Cradle-SCTetra	0.516	1.213	0.542
CSSRC-Fluent	0.506	1.223	0.526
HSVA-PPB	0.489	1.137	0.547
HSVA-QCM	0.470	1.160	0.516
INSEAN-PFC	0.479	1.129	0.541
MARIC-Fluent	0.503	1.170	0.548
SSPA-Fluent	0.499	1.182	0.537
SVA-Vortex	0.481	1.128	0.544
TUHH-Fresco+	0.501	1.203	0.530
UniGenua-PANEL	0.523	1.192	0.558
UniGenua-OpenFOAM	0.509	1.280	0.506
UniGenua-StarCCM(kw)	0.501	1.208	0.528
UniGenua-StarCCM(ke)	0.510	1.229	0.528
UniTriest	0.510	1.209	0.537
VicusDT-StarCCM	0.485	1.140	0.545
VOITH-Comet	0.490	1.193	0.523
VOITH-OpenFOAM	0.483	1.193	0.516
VTT-FinFlo	0.503	1.201	0.533
Measurement	0.510	1.178	0.551



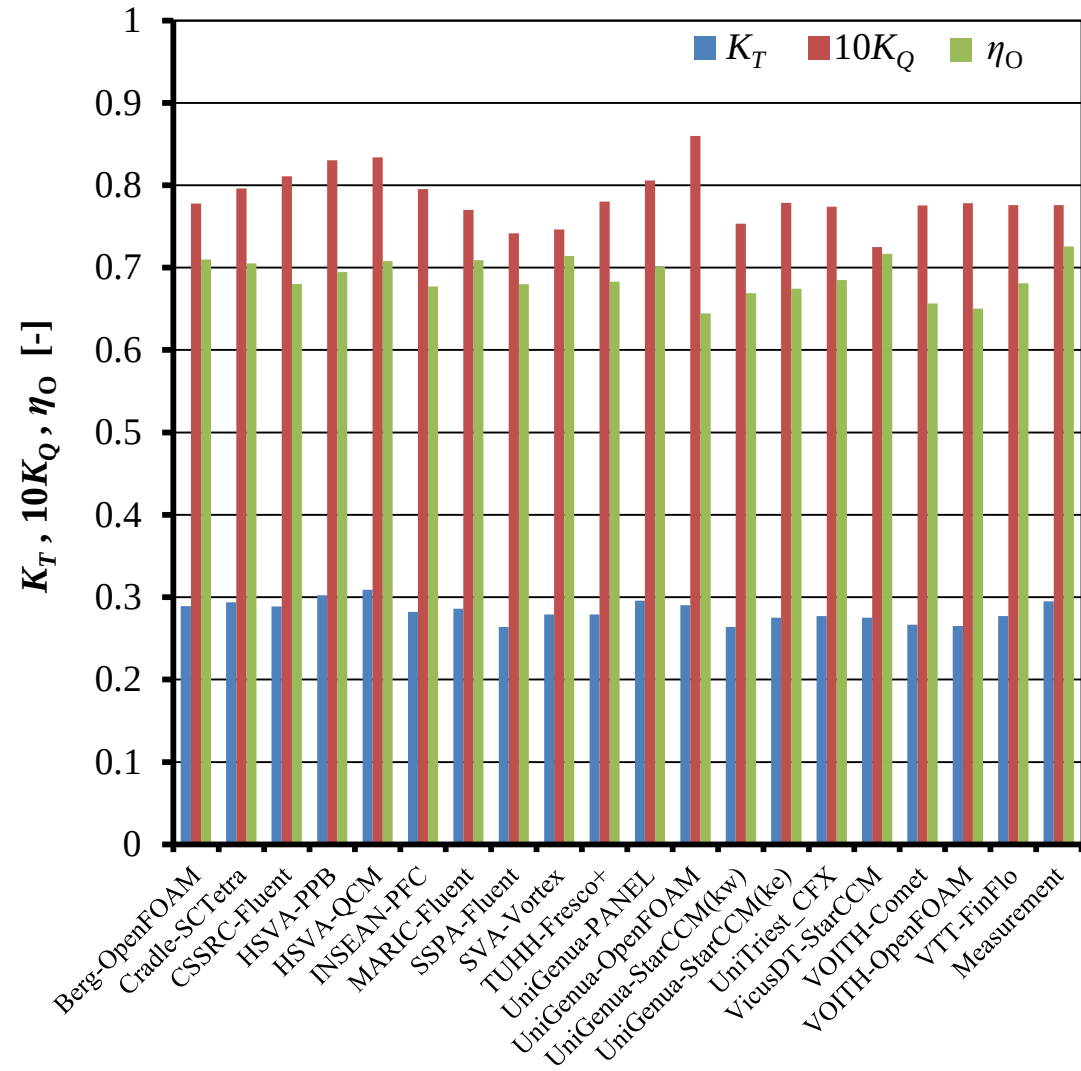
7.22 Comparison of open water characteristics, $J = 1.0$

	K_T [-]	$10K_Q$ [-]	η_o [-]
Berg-OpenFOAM	0.397	0.984	0.643
Cradle-SCTetra	0.405	1.012	0.636
CSSRC-Fluent	0.397	1.022	0.619
HSVA-PPB	0.397	1.001	0.631
HSVA-QCM	0.392	1.004	0.622
INSEAN-PFC	0.384	0.972	0.628
MARIC-Fluent	0.390	0.966	0.643
SSPA-Fluent	0.377	0.961	0.624
SVA-Vortex	0.383	0.941	0.648
TUHH-Fresco+	0.393	0.999	0.626
UniGenua-PANEL	0.410	1.013	0.643
UniGenua-OpenFOAM	0.404	1.084	0.593
UniGenua-StarCCM(kw)	0.377	0.974	0.616
UniGenua-StarCCM(ke)	0.387	0.996	0.618
UniTriest	0.390	0.992	0.626
VicusDT-StarCCM	0.380	0.915	0.659
VOITH-Comet	0.376	0.987	0.607
VOITH-OpenFOAM	0.373	0.988	0.600
VTT-FinFlo	0.386	0.989	0.622
Measurement	0.399	0.975	0.652



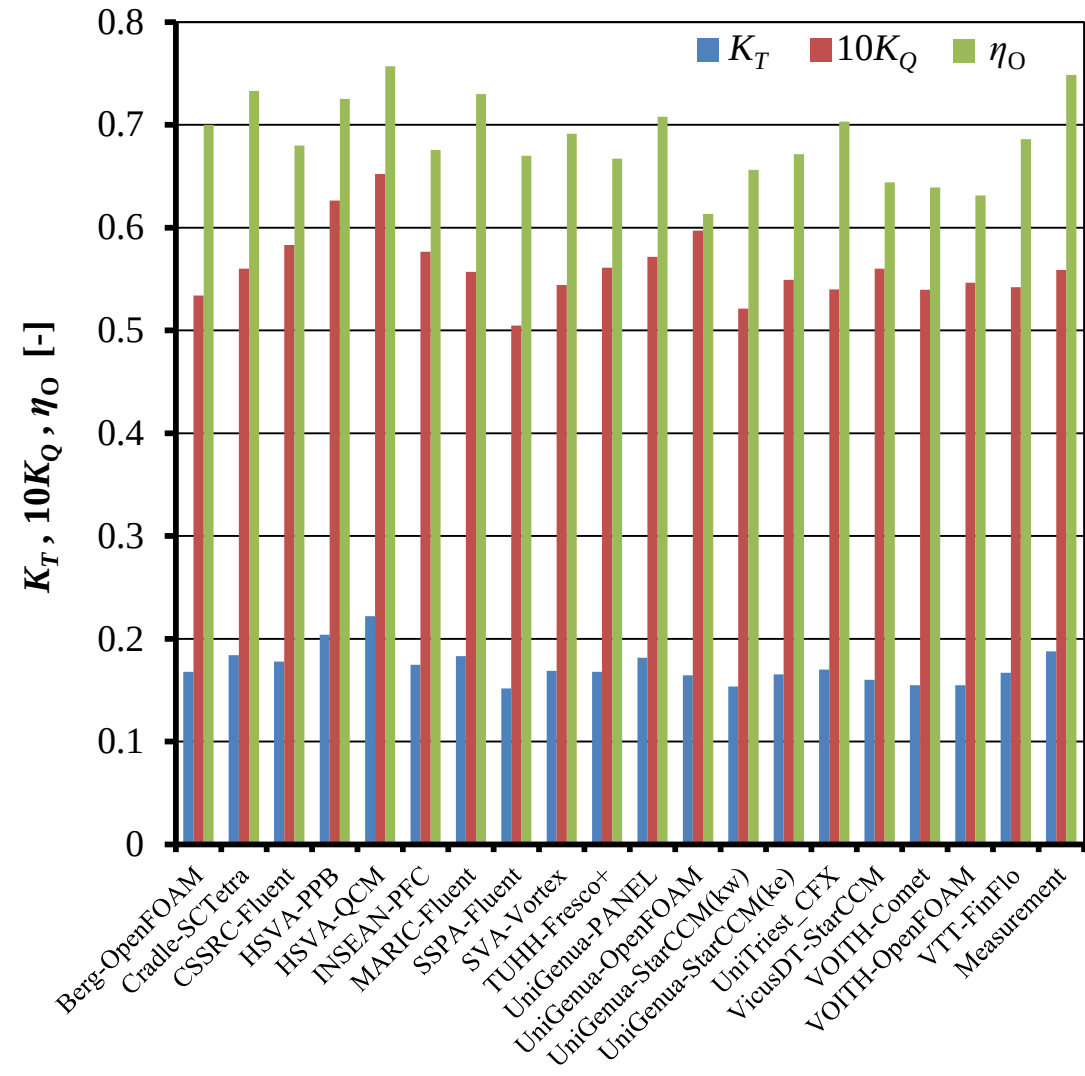
7.23 Comparison of open water characteristics, $J = 1.2$

	K_T [-]	$10K_Q$ [-]	η_o [-]
Berg-OpenFOAM	0.289	0.778	0.710
Cradle-SCTetra	0.294	0.796	0.705
CSSRC-Fluent	0.289	0.811	0.680
HSVA-PPB	0.302	0.831	0.695
HSVA-QCM	0.309	0.834	0.708
INSEAN-PFC	0.282	0.795	0.677
MARIC-Fluent	0.286	0.770	0.709
SSPA-Fluent	0.264	0.742	0.680
SVA-Vortex	0.279	0.746	0.714
TUHH-Fresco+	0.279	0.780	0.683
UniGenua-PANEL	0.296	0.806	0.701
UniGenua-OpenFOAM	0.290	0.860	0.644
UniGenua-StarCCM(kw)	0.264	0.754	0.669
UniGenua-StarCCM(ke)	0.275	0.778	0.675
UniTriest	0.277	0.774	0.685
VicusDT-StarCCM	0.275	0.725	0.717
VOITH-Comet	0.267	0.776	0.657
VOITH-OpenFOAM	0.265	0.778	0.650
VTT-FinFlo	0.277	0.776	0.681
Measurement	0.295	0.776	0.726



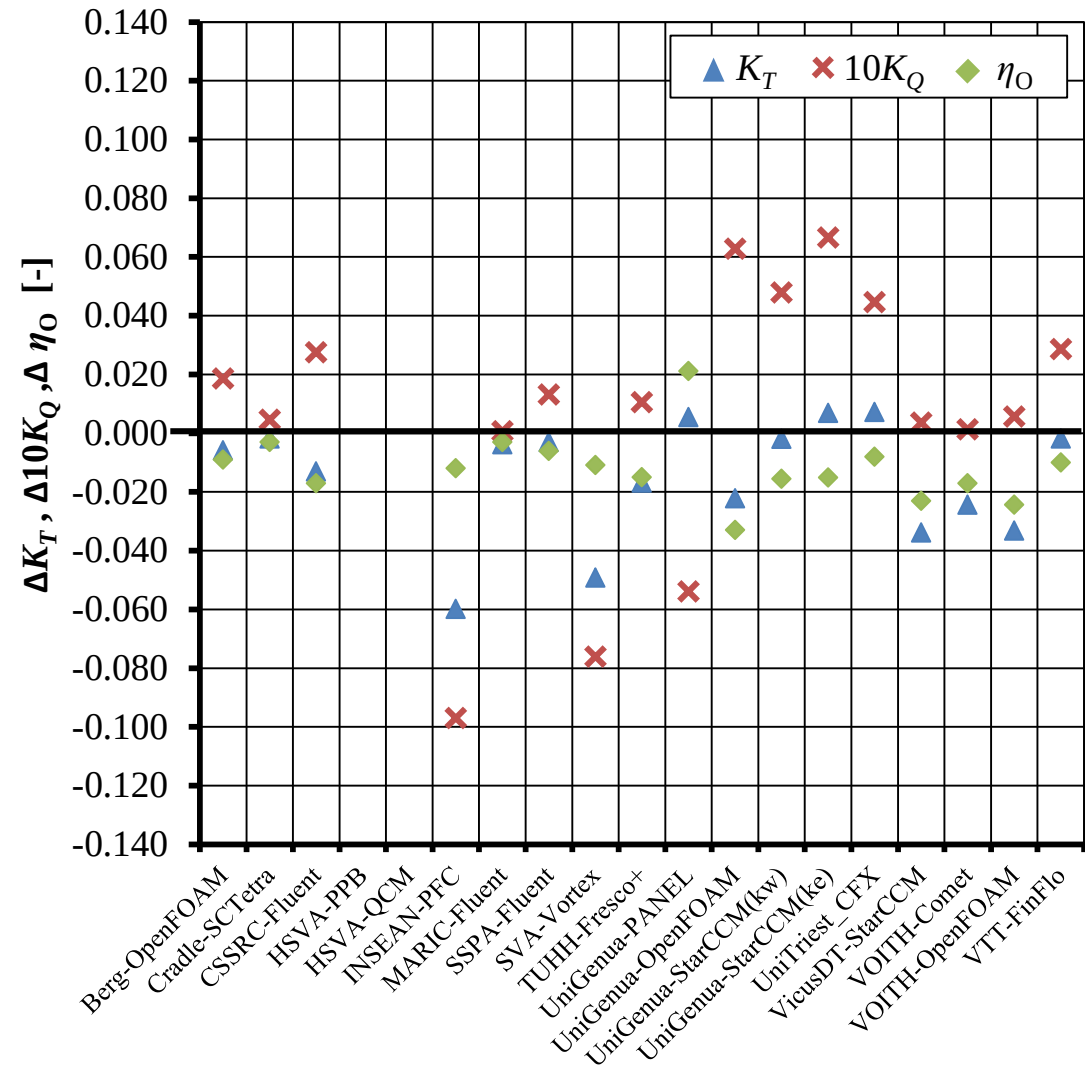
7.24 Comparison of open water characteristics, $J = 1.4$

	K_T [-]	$10K_Q$ [-]	η_o [-]
Berg-OpenFOAM	0.168	0.534	0.700
Cradle-SCTetra	0.184	0.560	0.733
CSSRC-Fluent	0.178	0.583	0.680
HSVA-PPB	0.204	0.626	0.725
HSVA-QCM	0.222	0.652	0.757
INSEAN-PFC	0.175	0.577	0.676
MARIC-Fluent	0.183	0.557	0.730
SSPA-Fluent	0.152	0.505	0.670
SVA-Vortex	0.169	0.544	0.691
TUHH-Fresco+	0.168	0.561	0.667
UniGenua-PANEL	0.182	0.572	0.708
UniGenua-OpenFOAM	0.164	0.597	0.613
UniGenua-StarCCM(kw)	0.153	0.521	0.656
UniGenua-StarCCM(ke)	0.165	0.549	0.671
UniTriest	0.170	0.540	0.703
VicusDT-StarCCM	0.160	0.560	0.644
VOITH-Comet	0.155	0.539	0.639
VOITH-OpenFOAM	0.155	0.546	0.631
VTT-FinFlo	0.167	0.542	0.686
Measurement	0.188	0.559	0.749



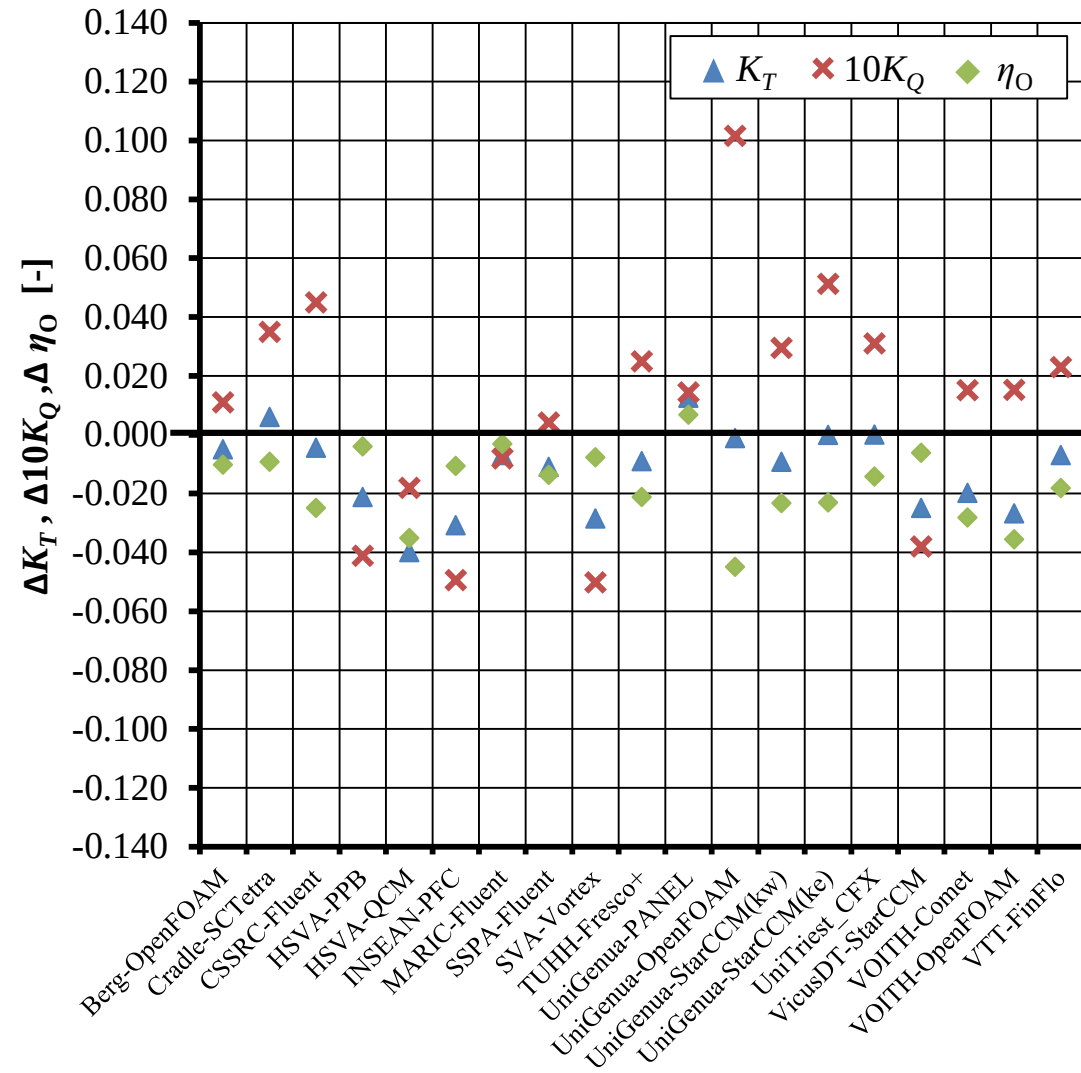
7.25 Absolute difference between measurement and calculation, $J = 0.6$

	ΔK_T [-]	$\Delta 10K_Q$ [-]	$\Delta \eta_O$ [-]
Berg-OpenFOAM	-0.006	0.019	-0.009
Cradle-SCTetra	-0.002	0.005	-0.003
CSSRC-Fluent	-0.013	0.028	-0.017
HSVA-PPB			
HSVA-QCM			
INSEAN-PFC	-0.060	-0.097	-0.012
MARIC-Fluent	-0.004	0.001	-0.003
SSPA-Fluent	-0.003	0.013	-0.006
SVA-Vortex	-0.049	-0.076	-0.011
TUHH-Fresco+	-0.017	0.011	-0.015
UniGenua-PANEL	0.005	-0.054	0.021
UniGenua-OpenFOAM	-0.022	0.063	-0.033
UniGenua-StarCCM(kw)	-0.002	0.048	-0.016
UniGenua-StarCCM(ke)	0.007	0.067	-0.015
UniTriest	0.007	0.045	-0.008
VicusDT-StarCCM	-0.034	0.004	-0.023
VOITH-Comet	-0.024	0.001	-0.017
VOITH-OpenFOAM	-0.033	0.006	-0.024
VTT-FinFlo	-0.002	0.029	-0.010



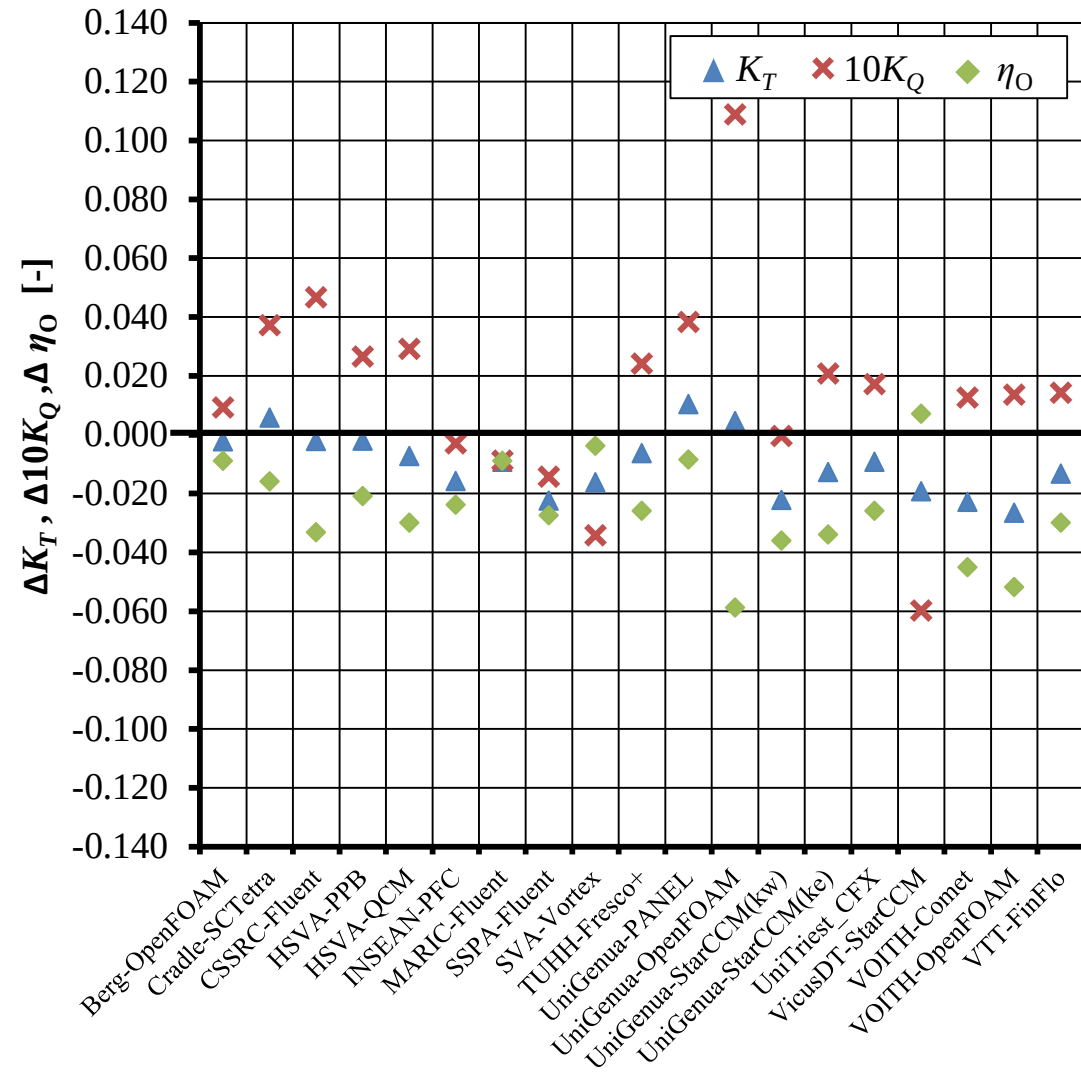
7.26 Absolute difference between measurement and calculation, $J = 0.8$

	ΔK_T [-]	$\Delta 10K_Q$ [-]	$\Delta \eta_O$ [-]
Berg-OpenFOAM	-0.005	0.011	-0.010
Cradle-SCTetra	0.006	0.035	-0.009
CSSRC-Fluent	-0.004	0.045	-0.025
HSVA-PPB	-0.021	-0.041	-0.004
HSVA-QCM	-0.040	-0.018	-0.035
INSEAN-PFC	-0.031	-0.049	-0.011
MARIC-Fluent	-0.007	-0.008	-0.003
SSPA-Fluent	-0.011	0.004	-0.014
SVA-Vortex	-0.029	-0.050	-0.008
TUHH-Fresco+	-0.009	0.025	-0.021
UniGenua-PANEL	0.013	0.014	0.007
UniGenua-OpenFOAM	-0.001	0.102	-0.045
UniGenua-StarCCM(kw)	-0.009	0.030	-0.023
UniGenua-StarCCM(ke)	0.000	0.051	-0.023
UniTriest	0.000	0.031	-0.014
VicusDT-StarCCM	-0.025	-0.038	-0.006
VOITH-Comet	-0.020	0.015	-0.028
VOITH-OpenFOAM	-0.027	0.015	-0.036
VTT-FinFlo	-0.007	0.023	-0.018



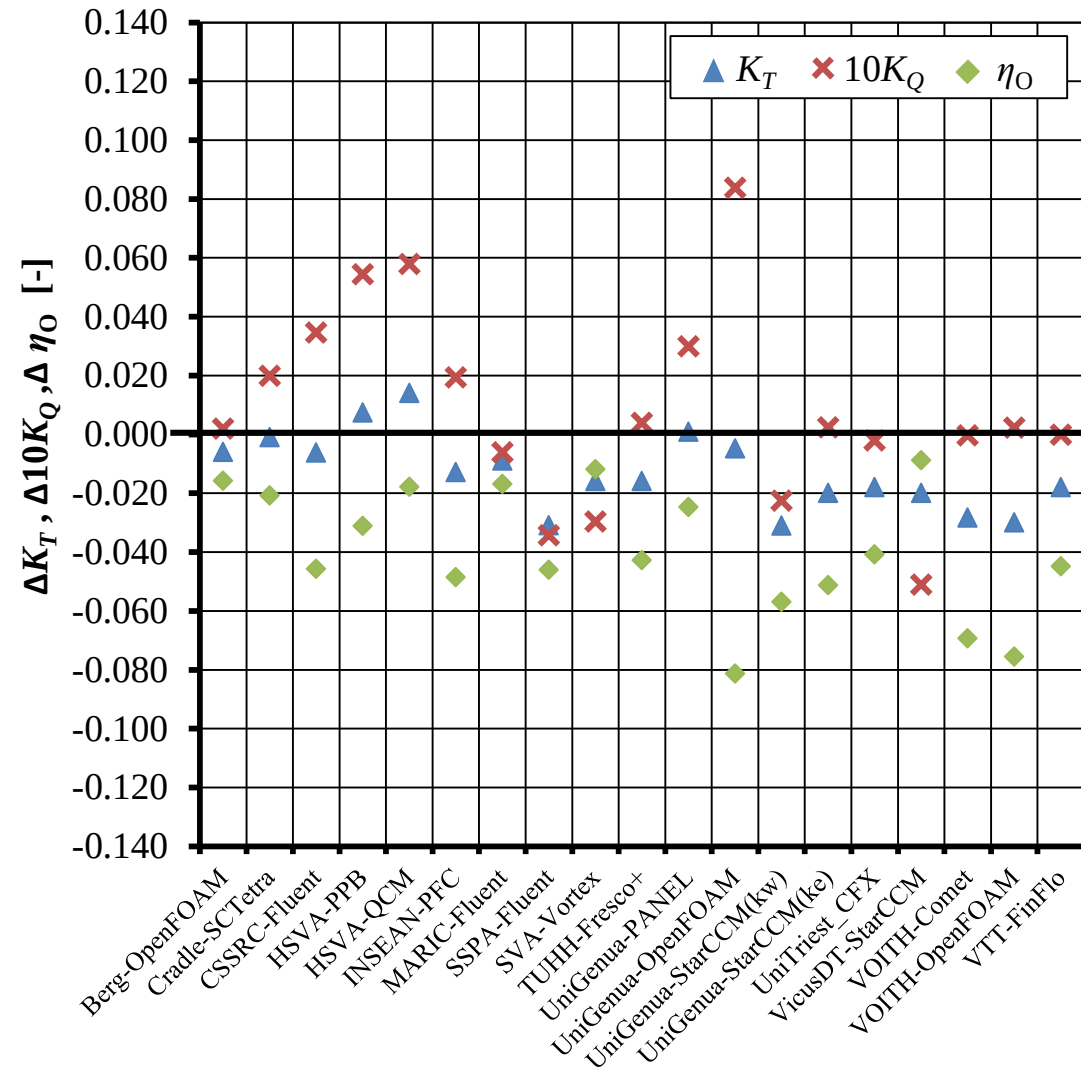
7.27 Absolute difference between measurement and calculation, $J = 1.0$

	ΔK_T [-]	$\Delta 10K_Q$ [-]	$\Delta \eta_O$ [-]
Berg-OpenFOAM	-0.002	0.009	-0.009
Cradle-SCTetra	0.006	0.037	-0.016
CSSRC-Fluent	-0.002	0.047	-0.033
HSVA-PPB	-0.002	0.026	-0.021
HSVA-QCM	-0.007	0.029	-0.030
INSEAN-PFC	-0.016	-0.003	-0.024
MARIC-Fluent	-0.009	-0.009	-0.009
SSPA-Fluent	-0.022	-0.014	-0.027
SVA-Vortex	-0.016	-0.034	-0.004
TUHH-Fresco+	-0.006	0.024	-0.026
UniGenua-PANEL	0.010	0.038	-0.009
UniGenua-OpenFOAM	0.004	0.109	-0.059
UniGenua-StarCCM(kw)	-0.022	-0.001	-0.036
UniGenua-StarCCM(ke)	-0.013	0.021	-0.034
UniTriest	-0.009	0.017	-0.026
VicusDT-StarCCM	-0.019	-0.060	0.007
VOITH-Comet	-0.023	0.013	-0.045
VOITH-OpenFOAM	-0.027	0.014	-0.052
VTT-FinFlo	-0.013	0.014	-0.030



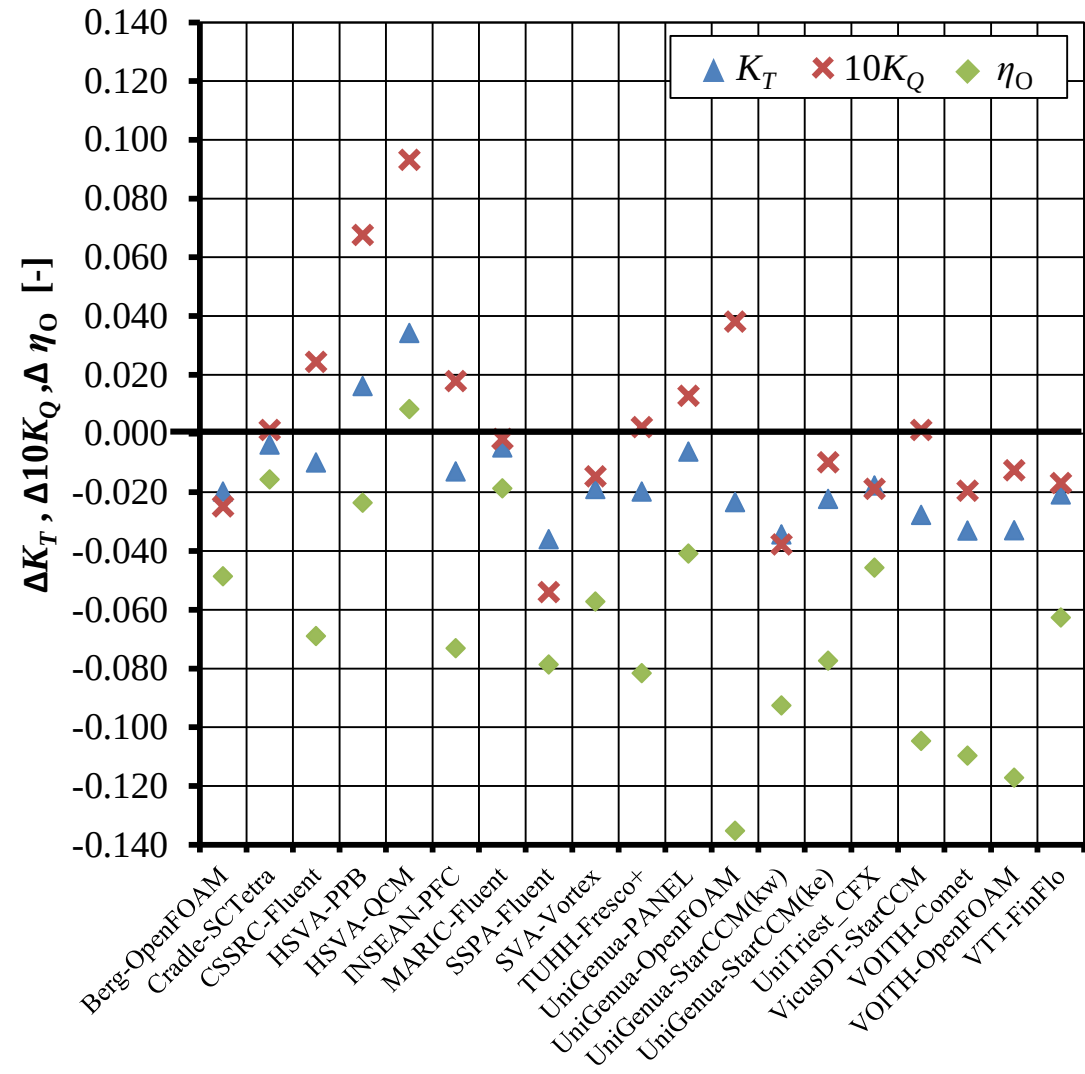
7.28 Absolute difference between measurement and calculation, $J = 1.2$

	ΔK_T [-]	$\Delta 10K_Q$ [-]	$\Delta \eta_O$ [-]
Berg-OpenFOAM	-0.006	0.002	-0.016
Cradle-SCTetra	-0.001	0.020	-0.021
CSSRC-Fluent	-0.006	0.035	-0.046
HSVA-PPB	0.007	0.054	-0.031
HSVA-QCM	0.014	0.058	-0.018
INSEAN-PFC	-0.013	0.019	-0.049
MARIC-Fluent	-0.009	-0.006	-0.017
SSPA-Fluent	-0.031	-0.034	-0.046
SVA-Vortex	-0.016	-0.030	-0.012
TUHH-Fresco+	-0.016	0.004	-0.043
UniGenua-PANEL	0.001	0.030	-0.025
UniGenua-OpenFOAM	-0.005	0.084	-0.081
UniGenua-StarCCM(kw)	-0.031	-0.023	-0.057
UniGenua-StarCCM(ke)	-0.020	0.002	-0.051
UniTriest	-0.018	-0.002	-0.041
VicusDT-StarCCM	-0.020	-0.051	-0.009
VOITH-Comet	-0.028	0.000	-0.069
VOITH-OpenFOAM	-0.030	0.002	-0.076
VTT-FinFlo	-0.018	0.000	-0.045



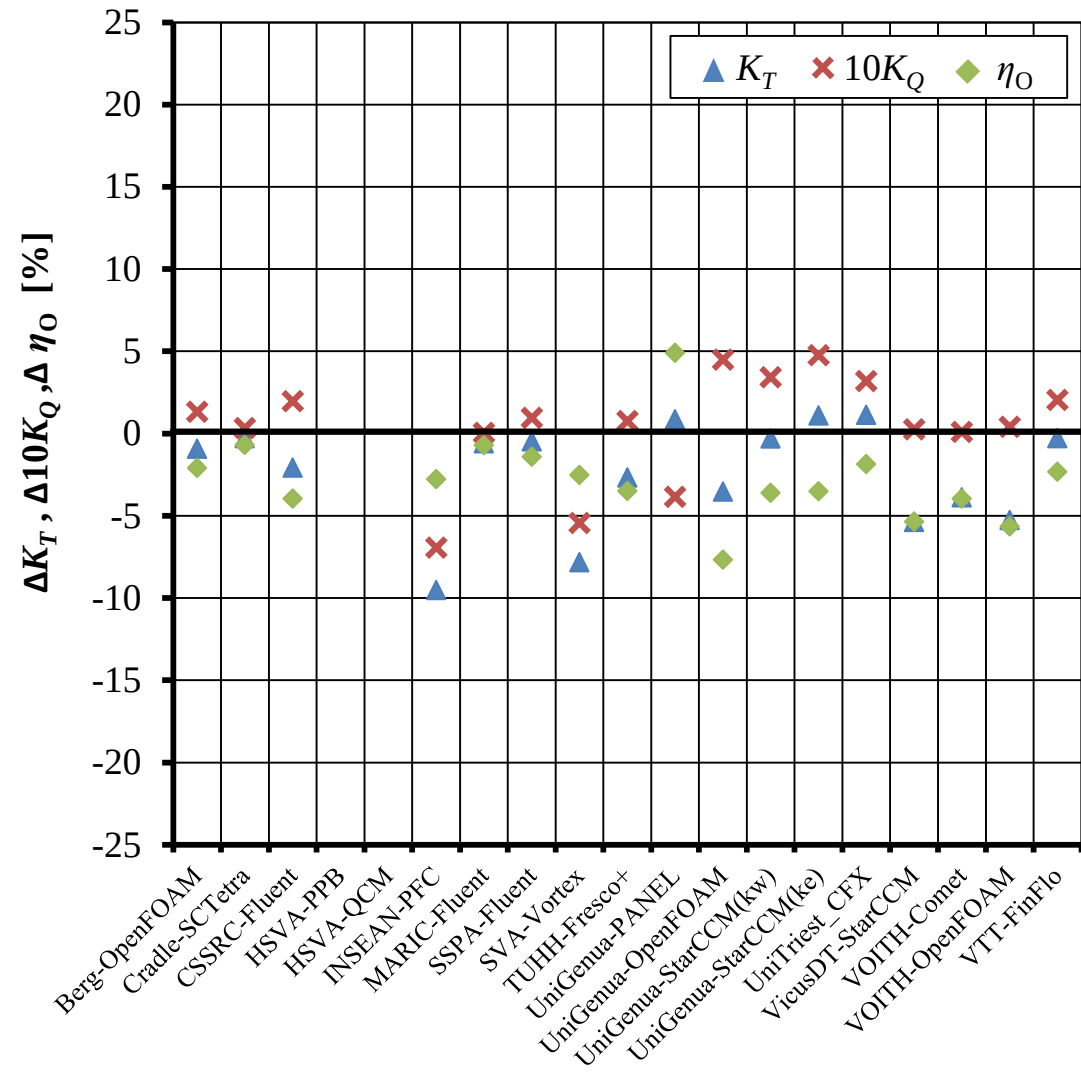
7.29 Absolute difference between measurement and calculation, $J = 1.4$

	ΔK_T [-]	$\Delta 10K_Q$ [-]	$\Delta \eta_O$ [-]
Berg-OpenFOAM	-0.020	-0.025	-0.049
Cradle-SCTetra	-0.004	0.001	-0.016
CSSRC-Fluent	-0.010	0.024	-0.069
HSVA-PPB	0.016	0.068	-0.024
HSVA-QCM	0.034	0.093	0.008
INSEAN-PFC	-0.013	0.018	-0.073
MARIC-Fluent	-0.005	-0.002	-0.019
SSPA-Fluent	-0.036	-0.054	-0.079
SVA-Vortex	-0.019	-0.015	-0.057
TUHH-Fresco+	-0.020	0.002	-0.082
UniGenua-PANEL	-0.006	0.013	-0.041
UniGenua-OpenFOAM	-0.023	0.038	-0.135
UniGenua-StarCCM(kw)	-0.034	-0.038	-0.093
UniGenua-StarCCM(ke)	-0.022	-0.010	-0.077
UniTriest	-0.018	-0.019	-0.046
VicusDT-StarCCM	-0.028	0.001	-0.105
VOITH-Comet	-0.033	-0.019	-0.110
VOITH-OpenFOAM	-0.033	-0.013	-0.117
VTT-FinFlo	-0.021	-0.017	-0.063



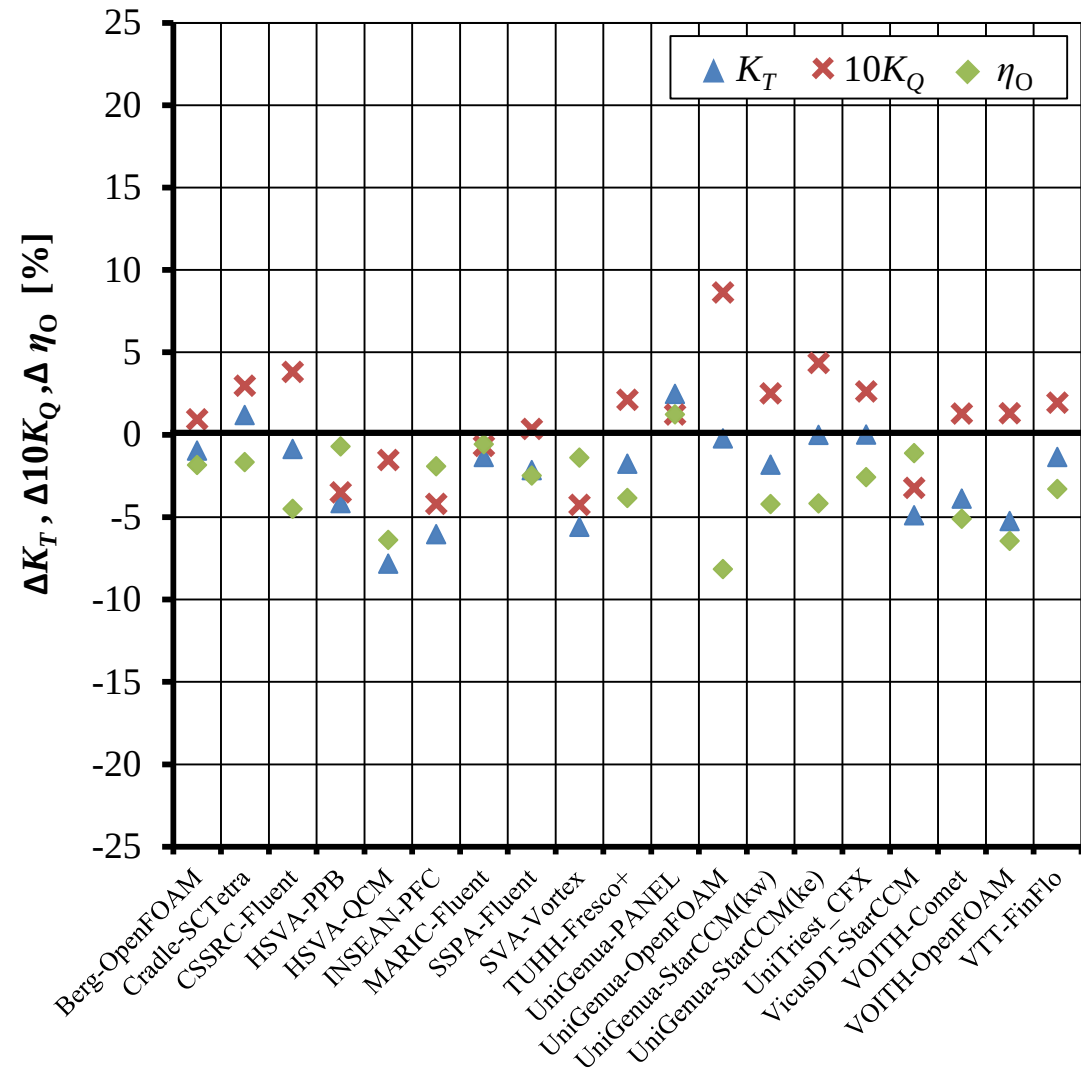
7.30 Relative difference between measurement and calculation, $J = 0.6$

	ΔK_T [%]	$\Delta 10K_Q$ [%]	$\Delta \eta_O$ [%]
Berg-OpenFOAM	-0.927	1.334	-2.101
Cradle-SCTetra	-0.291	0.331	-0.706
CSSRC-Fluent	-2.072	1.971	-3.961
HSVA-PPB			
HSVA-QCM			
INSEAN-PFC	-9.521	-6.944	-2.770
MARIC-Fluent	-0.609	0.045	-0.711
SSPA-Fluent	-0.477	0.947	-1.411
SVA-Vortex	-7.824	-5.448	-2.515
TUHH-Fresco+	-2.676	0.761	-3.496
UniGenua-PANEL	0.864	-3.854	4.907
UniGenua-OpenFOAM	-3.526	4.496	-7.677
UniGenua-StarCCM(kw)	-0.314	3.431	-3.620
UniGenua-StarCCM(ke)	1.090	4.775	-3.517
UniTriest	1.140	3.196	-1.868
VicusDT-StarCCM	-5.380	0.260	-5.356
VOITH-Comet	-3.879	0.095	-3.970
VOITH-OpenFOAM	-5.268	0.410	-5.659
VTT-FinFlo	-0.291	2.050	-2.333



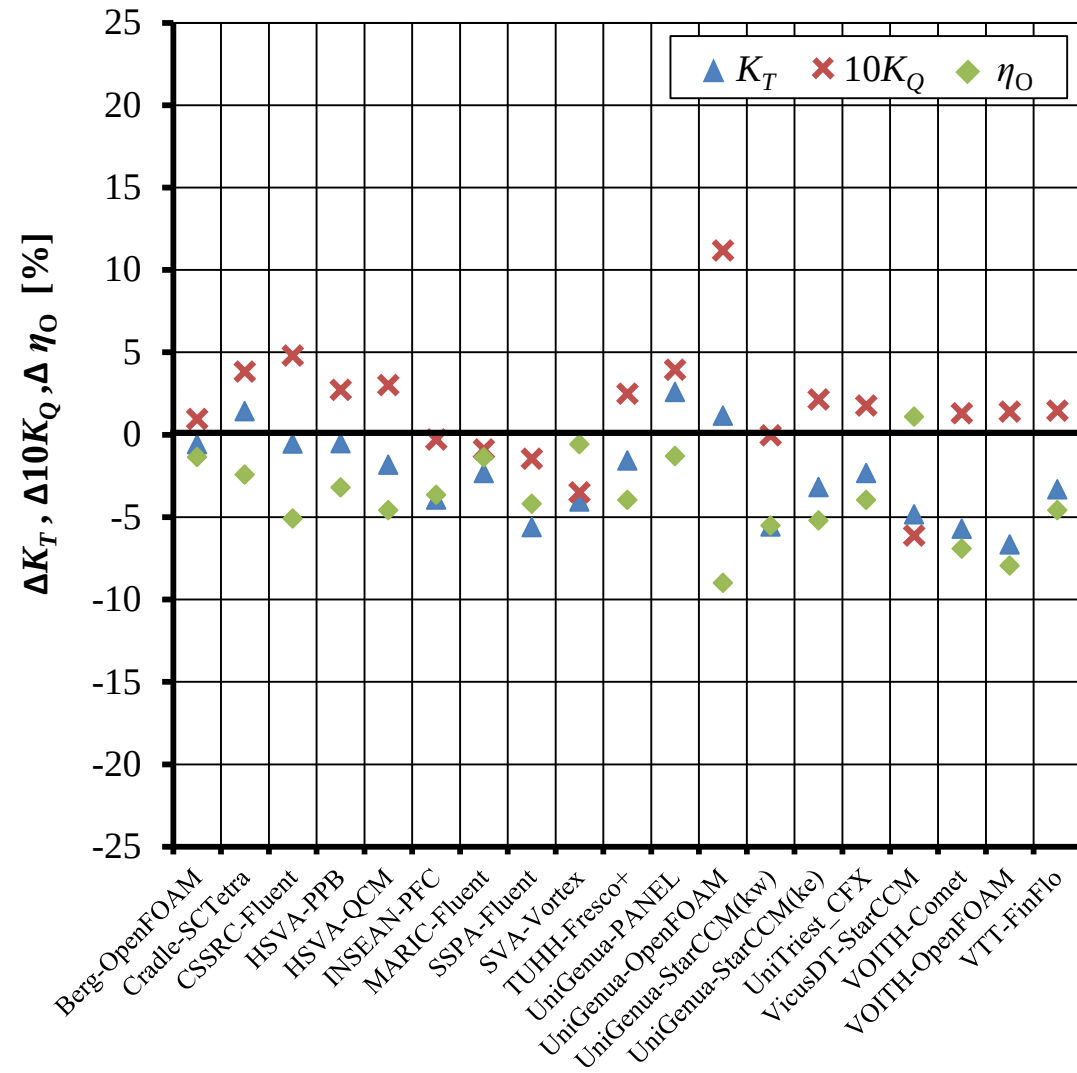
7.31 Relative difference between measurement and calculation, $J = 0.8$

	ΔK_T [%]	$\Delta 10K_Q$ [%]	$\Delta \eta_O$ [%]
Berg-OpenFOAM	-0.974	0.933	-1.849
Cradle-SCTetra	1.183	2.971	-1.668
CSSRC-Fluent	-0.876	3.819	-4.516
HSVA-PPB	-4.150	-3.490	-0.724
HSVA-QCM	-7.837	-1.529	-6.385
INSEAN-PFC	-6.043	-4.196	-1.928
MARIC-Fluent	-1.366	-0.680	-0.582
SSPA-Fluent	-2.150	0.352	-2.493
SVA-Vortex	-5.597	-4.262	-1.392
TUHH-Fresco+	-1.758	2.122	-3.845
UniGenua-PANEL	2.467	1.222	1.230
UniGenua-OpenFOAM	-0.230	8.626	-8.152
UniGenua-StarCCM(kw)	-1.820	2.509	-4.223
UniGenua-StarCCM(ke)	-0.009	4.350	-4.178
UniTriest	0.007	2.631	-2.575
VicusDT-StarCCM	-4.895	-3.226	-1.123
VOITH-Comet	-3.889	1.290	-5.113
VOITH-OpenFOAM	-5.248	1.298	-6.457
VTT-FinFlo	-1.366	1.952	-3.300



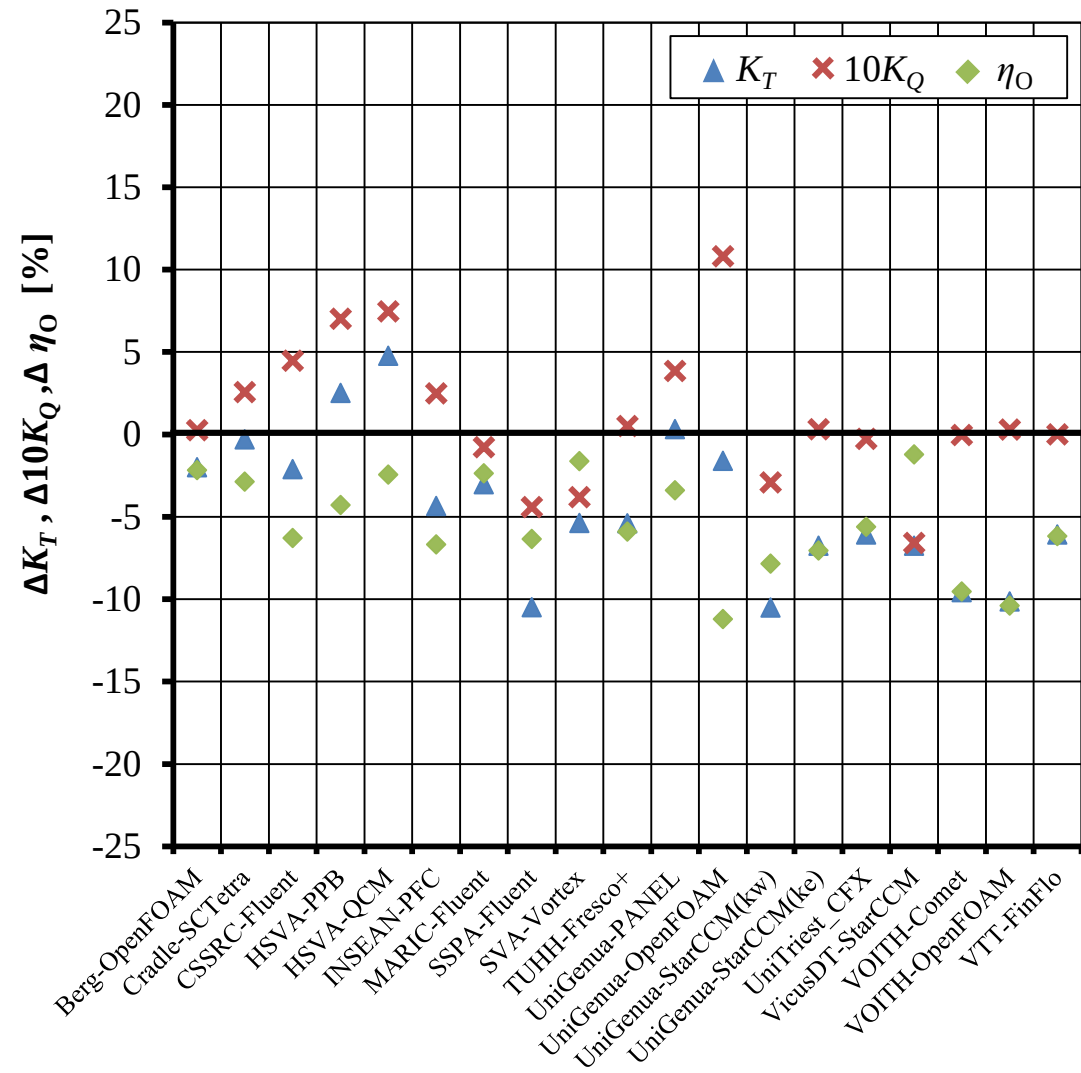
7.32 Relative difference between measurement and calculation, $J = 1.0$

	ΔK_T [%]	$\Delta 10K_Q$ [%]	$\Delta \eta_O$ [%]
Berg-OpenFOAM	-0.594	0.936	-1.380
Cradle-SCTetra	1.409	3.808	-2.454
CSSRC-Fluent	-0.569	4.782	-5.107
HSVA-PPB	-0.544	2.710	-3.220
HSVA-QCM	-1.846	2.987	-4.601
INSEAN-PFC	-3.969	-0.312	-3.669
MARIC-Fluent	-2.346	-0.911	-1.399
SSPA-Fluent	-5.627	-1.471	-4.218
SVA-Vortex	-4.084	-3.511	-0.593
TUHH-Fresco+	-1.595	2.474	-3.987
UniGenua-PANEL	2.561	3.923	-1.310
UniGenua-OpenFOAM	1.127	11.158	-9.026
UniGenua-StarCCM(kw)	-5.595	-0.056	-5.542
UniGenua-StarCCM(ke)	-3.210	2.128	-5.227
UniTriest	-2.346	1.756	-3.987
VicusDT-StarCCM	-4.850	-6.142	1.074
VOITH-Comet	-5.737	1.284	-6.929
VOITH-OpenFOAM	-6.678	1.387	-7.960
VTT-FinFlo	-3.348	1.448	-4.601



7.33 Relative difference between measurement and calculation, $J = 1.2$

	ΔK_T [%]	$\Delta 10K_Q$ [%]	$\Delta \eta_O$ [%]
Berg-OpenFOAM	-2.008	0.251	-2.177
Cradle-SCTetra	-0.312	2.571	-2.866
CSSRC-Fluent	-2.109	4.465	-6.297
HSVA-PPB	2.502	7.017	-4.285
HSVA-QCM	4.774	7.468	-2.453
INSEAN-PFC	-4.354	2.496	-6.683
MARIC-Fluent	-3.025	-0.779	-2.370
SSPA-Fluent	-10.485	-4.417	-6.348
SVA-Vortex	-5.382	-3.810	-1.633
TUHH-Fresco+	-5.398	0.509	-5.897
UniGenua-PANEL	0.317	3.854	-3.406
UniGenua-OpenFOAM	-1.611	10.806	-11.207
UniGenua-StarCCM(kw)	-10.520	-2.903	-7.845
UniGenua-StarCCM(ke)	-6.766	0.310	-7.054
UniTriest	-6.077	-0.264	-5.622
VicusDT-StarCCM	-6.755	-6.578	-1.213
VOITH-Comet	-9.583	-0.045	-9.547
VOITH-OpenFOAM	-10.145	0.290	-10.403
VTT-FinFlo	-6.077	-0.006	-6.173



7.34 Relative difference between measurement and calculation, $J = 1.4$

	ΔK_T [%]	$\Delta 10K_Q$ [%]	$\Delta \eta_O$ [%]
Berg-OpenFOAM	-10.525	-4.442	-6.499
Cradle-SCTetra	-2.004	0.211	-2.092
CSSRC-Fluent	-5.253	4.362	-9.211
HSVA-PPB	8.594	12.093	-3.147
HSVA-QCM	18.234	16.674	1.114
INSEAN-PFC	-6.886	3.197	-9.770
MARIC-Fluent	-2.537	-0.326	-2.556
SSPA-Fluent	-19.153	-9.656	-10.512
SVA-Vortex	-10.067	-2.615	-7.652
TUHH-Fresco+	-10.525	0.390	-10.907
UniGenua-PANEL	-3.296	2.293	-5.464
UniGenua-OpenFOAM	-12.475	6.826	-18.070
UniGenua-StarCCM(kw)	-18.289	-6.759	-12.366
UniGenua-StarCCM(ke)	-11.895	-1.746	-10.330
UniTriest	-9.460	-3.368	-6.099
VicusDT-StarCCM	-14.786	0.211	-13.980
VOITH-Comet	-17.619	-3.476	-14.653
VOITH-OpenFOAM	-17.556	-2.241	-15.663
VTT-FinFlo	-11.058	-3.010	-8.370

